

Algorithmic Ranking and Delivery Foreclosure in Retail Platforms: Evidence from Amazon and Mercado Libre

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September 19, 2026

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Abstract

Retail platforms control product rankings and often vertically integrate into delivery services, such as Fulfilled by Amazon, creating an incentive to favor sellers who adopt them. The FTC and European Commission are investigating this practice, but its welfare effects remain unquantified. Using a panel of more than 415,000 observations from Amazon and Mercado Libre, Mexico’s two largest retail platforms, I estimate a structural model of consumer platform choice, search, seller pricing and delivery choices, and platform ranking. I find that integrated delivery is higher quality than third-party alternatives but up to 84% more expensive, and receives systematically better rankings. Barring platforms from favoring their integrated delivery in rankings reduces its adoption by 5%, lowers prices by up to 7%, and raises total surplus by 4.5%, with Amazon’s consumer surplus rising 12% against only 2% for Mercado Libre. Separating Mercado Libre’s delivery service from the platform lowers total surplus by 4.4% and consumer surplus on Amazon by 10%, showing that rival-platform competition disciplines self-preferencing even without regulation.

Keywords: algorithmic self-preferencing, platform competition, vertical integration, search frictions, e-commerce, antitrust

*Carnegie Mellon University. gbasulto@andrew.cmu.edu. I am deeply grateful to my PhD advisor, Aniko Öry, for her invaluable guidance and the generous time she has dedicated to discussing this work throughout the course of this project. I also thank my committee members Anh Nguyen and Maryam Saeedi for their thoughtful comments and continued support, and my dissertation external readers Alan Montgomery and Minkyung Kim for their engagement with this work and their helpful feedback. All remaining errors are my own.

1 Introduction

Retail platforms act as intermediaries between third-party sellers and consumers, primarily through ranking algorithms that determine which products appear (and in what order) in response to a consumer’s search. In addition to this intermediation role, many platforms vertically integrate into complementary services, such as warehousing and delivery, e.g. *Fulfilled by Amazon (FBA)*. The welfare implications of this vertical integration are ambiguous. On the one hand, platforms may use their ranking algorithms to favor sellers that adopt the platform’s integrated services, reducing the visibility of alternative options for consumers and increasing the platform’s pricing power in the delivery market. On the other hand, integrated delivery services may genuinely improve efficiency and consumer experience, potentially justifying both their prominence in search results and their higher cost.

In this paper, I examine these forces empirically using collected data from the two largest vertically integrated retail platforms in Mexico: Amazon (FBA) and Mercado Libre (Full). I show that products using the platforms’ integrated delivery services receive systematically better rankings in search results, with Full moving products up about 6 positions on Mercado Libre and FBA moving products up about 4 positions on Amazon. I find that integrated delivery services are of higher quality than third-party alternatives. They are also up to 84% more expensive. When platforms are barred from favoring products that use their own delivery services in rankings, sellers reduce their use of these services by up to 5%, prices fall by 3-7%, and total welfare increases by 4.56%. The effects are highly asymmetric across platforms: because Mercado Libre’s baseline algorithm favors integrated delivery more heavily than Amazon’s, regulating rankings induces a larger adjustment in Mercado Libre’s search results and shifts consumer market share toward Amazon. I then use a second counterfactual to show that competition between the platforms also constrains this market power. When Mercado Libre’s Full service is separated from the platform while Amazon’s ranking algorithm and vertical relation with FBA are left unchanged, total surplus falls by 4.43% and Amazon consumer surplus falls by 10.56%. These results show that platforms use their matching role to over-promote their own delivery services and extract rents from sellers, while competition from a rival platform lowers a platform ability to self-preference.

This paper speaks directly to important recent policy concerns. In September 2023, the U.S. Federal Trade Commission sued Amazon, alleging that tying Fulfillment by Amazon (FBA) to Prime eligibility constitutes an illegal monopolization scheme that raises costs for sellers and ultimately harms consumers through higher prices (Federal Trade Commission, 2023). Similarly, the European Commission’s 2019–2022 investigation into Amazon concluded that the platform’s ranking algorithms systematically favored FBA products in the Buy Box, prompting binding commitments that Amazon treat all sellers equally regardless of fulfillment method (European Commission, 2022). Closest to this study’s empirical setting, Mexico’s competition authority COFECE issued a preliminary report in February 2024 documenting that Amazon and Mercado Libre, which together control over 35% of online marketplace sales in Mexico, create barriers to competition by giving “preferential treatment to products from sellers who use their fulfillment services” and maintaining opaque ranking algorithms that disad-

vantage third-party logistics providers (COFECE, 2024).

Despite this regulatory attention, quantifying the welfare effects of platform-delivery integration has proven difficult. Enforcement actions to date rely primarily on qualitative evidence and theoretical models of foreclosure, which leaves open questions about magnitudes: how much of the harm from ranking bias falls on sellers using rival delivery providers versus on consumers through higher prices, and how much market power platforms gain from this practice relative to one another. The academic literature has not filled this gap either, for two reasons. First, it has focused on a different mechanism than the one at stake in these cases. Second, it has almost always studied a single platform, which limits what it can say about competition between platforms.

On the first point, most existing evidence concerns platforms self-preferencing their *competing products*, for example Amazon promoting its private-label brands or sponsored listings, where foreclosure works through product-market substitution: consumers are steered away from rival products toward the platform’s own line (Zhu and Liu, 2018, Farronato et al., 2023, Dubé, 2022). The mechanism studied here is different since the platforms might favor their own delivery service, not their competing products. Foreclosure in this setting operates in the delivery market rather than the product market, as third-party logistics providers lose business when sellers switch to more expensive platform-integrated delivery to gain visibility. The welfare cost of this practice shows up as higher delivery fees passed through into consumer prices on top of reduced product variety. Closest to this part of the paper, Raval (2022) documents that Amazon’s ranking algorithm favors products using its own fulfillment arm even more strongly than products sold directly by Amazon, which is reduced-form evidence of the kind of delivery self-preferencing this paper studies. My paper builds on this finding by developing an equilibrium model of platform ranking, seller delivery adoption, pricing, and consumer search, which lets me quantify how much of the resulting harm falls on sellers using third-party delivery versus consumer facing prices higher prices, and how much a regulatory change to the ranking algorithm would move each margin.

On the second point, almost all of the existing evidence on algorithmic bias, from Zhu and Liu (2018) and Farronato et al. (2023) to recent structural work (Musolf and Kwok Hao, n.d., Hartzell and Haupt, 2024, Lam, 2021, Chen and Tsai, 2024), studies Amazon by itself. This is a natural starting point given data availability and Amazon’s position as the largest online marketplace in both the U.S. and Europe, but it means these papers can speak to how ranking bias redistributes surplus between a platform and its own sellers, and not to how it shapes competition between platforms. Because I observe Amazon and Mercado Libre through the same structural model, I can show that the platform with the more biased ranking algorithm extracts more rents from its sellers in the baseline but also stands to lose more market share once both platforms are held to a common ranking standard. I can also ask how the conduct of one platform changes when the competitive constraint imposed by the other platform is weakened. This provides a statement about competition in the market for platforms that a single-platform design cannot make. It connects my results to a theoretical literature on platform competition (Armstrong, 2006, Rochet and Tirole, 2003, Armstrong and Wright, 2007) that has, to my knowledge, rarely been confronted with data from

platforms that actually compete for the same consumers and sellers.

These two gaps mean that although the COFECE, FTC, and European Commission cases all turn on questions of magnitude, no existing paper provides an equilibrium estimate of how much delivery-based ranking bias forecloses third-party providers, raises prices, or shifts market power between competing platforms, nor how much competition between platforms constrains this conduct. This paper provides structural estimates addressing these questions in a setting directly relevant to ongoing enforcement, allowing the counterfactual simulations to speak both to remedies that directly regulate ranking algorithms and to the role of platform competition in limiting self-preferencing.

To do this, I construct a panel dataset of product-seller observations from the two largest retail platforms in Mexico, Amazon and Mercado Libre, documenting the pricing and usage of their integrated delivery services, *Fulfilled by Amazon* and *Full*. I collected data from Amazon and Mercado Libre search results for specific keywords over the period from May 2023 to May 2025 using web scraping. The search results include product prices, monthly sales buckets, and rankings, along with all relevant information about the delivery method. I supplement this with data on the referral fees charged by the platforms, as well as the warehousing and delivery costs associated with FBA and Full, which I matched to the observed products. Finally, I incorporate aggregate data on the platforms’ market shares and market sizes across different subcategories of the online retail space from industry reports. The main dataset used in this paper consists of over 415,000 product-seller-time observations across various subcategories (e.g., electronics, men’s clothing) from both Amazon and Mercado Libre, including 12,950 unique seller-product pairs.

I estimate a structural model of consumers’ choice across platforms based on platform inherent quality as well as expectations of the products offered, consumer search with frictions, the platforms’ ranking and delivery pricing decisions, and seller pricing and delivery decisions. The estimation results reveal substantial search frictions on both platforms. On Amazon, 62% of consumers consider only the first 20 products in search results, while on Mercado Libre, 22% of consumers stop searching within the first 20 products and the median search depth is around 50 products. While consumers are willing to pay 25% extra for products using FBA or Full, I also find evidence of algorithmic bias favoring integrated delivery. Products using FBA or Full receive systematically better rankings, moving up about 4 positions on Amazon and 6 positions on Mercado Libre relative to comparable products using third-party delivery, even after controlling for price, sales history, and reviews. This preferential treatment comes at a cost: platform delivery services are about 84% more expensive than third-party alternatives. I also document that, given two similar products, the one with the higher price is more likely to appear in the better ranking, so the algorithms reduce the visibility both of third-party delivery and of cheaper product alternatives.

I then use the estimated model to conduct two sets of counterfactual exercises. The first asks what happens if both platforms are barred from favoring products that use their own integrated delivery service in rankings. I consider the adjustment under three progressively less restrictive equilibrium concepts. First, I hold seller prices and deliv-

ery choices fixed to isolate the direct matching effect of the ranking change. Consumer surplus increases slightly in the two platforms (0.62% on Mercado Libre and 1.96% on Amazon) as consumers face better choice sets. Second, I allow sellers to reoptimize prices and delivery choices. Since using integrated delivery does not lead to higher visibility on the counterfactual its use decreases by around 5% on both platforms and prices drop by 3% on Mercado Libre and 7% on Amazon. These first two equilibria assume that the platforms do not respond to the interventions so they can be read as benchmark cases to understand and measure the effect of each force in the model. However, platforms might respond to the new rankings by adjusting other parts of their business, which could dampen the intended effects of the policy. So, in the third step I allow the platforms to adjust their commission rates under the new rankings. In the platform equilibrium, removing delivery-based ranking preferences increases total surplus by 4.56%. Integrated delivery adoption falls by 5.42% on Amazon and 4.79% on Mercado Libre as sellers still lose the visibility incentive to use the more expensive integrated service. Prices fall by less than in the seller equilibrium (7.65% on Amazon and 2.27% on Mercado Libre) since platforms increase their commission rate which rises sellers costs. Consumer surplus rises on both platforms, but the effects are strongly asymmetric: it increases by 12.09% on Amazon and only 2.10% on Mercado Libre. The asymmetry is also visible in platform competition. Amazon’s consumer market share rises by 18.60%, while Mercado Libre’s falls by 14.34%. Mercado Libre responds to the loss of delivery rents by raising its commission rate substantially, which offsets part of the consumer gain. Because Mercado Libre’s baseline algorithm favors Full more aggressively than Amazon’s favors FBA, a common regulatory standard requires a larger adjustment in Mercado Libre’s rankings and shifts rents and consumers toward the less-biased platform. Thus, a formally neutral ranking rule has important distributional effects across competing platforms.

Some features of the model limit how the counterfactual results should be read. First, I assume that both sides of the market single-home: each consumer searches and buys on only one platform, and each seller lists its products on only one platform. The low overlap of products across platforms (Section 3) supports this assumption for sellers, but some consumers likely search on both Amazon and Mercado Libre before buying. A consumer who multihomes can get around a biased ranking on one platform by finding a cheaper or better-matched product on the other. Multihoming would therefore limit the harm from ranking bias in the baseline, and my estimates of the consumer surplus gains from ranking regulation are best read as an upper bound. Second, I hold the set of sellers and products on each platform fixed and do not model entry or exit. This assumption can bias the results in either direction. A fairer ranking algorithm could attract new sellers, increase price competition, and raise consumer surplus beyond what I report. On the other hand, if platforms respond to the regulation by raising commission rates to recover lost delivery revenue, some sellers may exit, leaving consumers with fewer options and lowering consumer surplus. Since the model does not capture either force, the net effect of entry and exit on my welfare estimates is ambiguous.

The second counterfactual asks how much platform competition itself disciplines self-preferencing. In this exercise, I separate Mercado Libre’s Full service from the platform and assume that all sellers on Mercado Libre use third-party delivery ser-

vices at the estimated costs and thus remove the ranking advantage from using Full, while leaving Amazon’s ranking algorithm and FBA service unchanged. This exercise is not intended as a literal divestiture proposal. Instead, it approximates what happens to Amazon when the competitive constraint imposed by a rival delivery alternative is weakened. Mercado Libre’s own consumers gain substantially from cheaper delivery, with consumer surplus rising 7.73%, but Mercado Libre loses market share and both producer and platform surplus go down. Amazon gains 16.92% in market share and 38.79% in platform surplus, yet Amazon consumer surplus falls by 10.56%. Total surplus falls by 4.43%. The comparison shows that the presence of a competing platform disciplines the exercise of market power even without direct regulation. The welfare value of this competitive constraint is therefore of a magnitude comparable to the direct welfare effect of regulating delivery-based ranking bias.

The counterfactuals distinguish two margins that are difficult to study in a single-platform setting. Direct ranking regulation limits the ability of vertically integrated platforms to charge sellers indirectly for visibility through costly fulfillment, while platform competition limits how aggressively a platform can exercise that market power in equilibrium. The results therefore suggest that competition authorities evaluating self-preferencing remedies should consider not only how a platform ranks products, but also how the conduct of competing platforms constrains those ranking incentives.

1.1 Literature Review

This paper connects to five strands of literature. The first talks about the economics of platform design and platform competition. A foundational line of work examines how platform design choices, including ranking mechanisms, fee structures, and advertising, affect the welfare of consumers and sellers. Choi and Mela (2019) highlight the tension between advertising and transaction fee revenue streams, showing that product ranking and ad pricing mechanisms jointly influence user experience and platform profitability. Long et al. (2022) show that platforms can strategically use information obtained from ad auctions to refine organic rankings and set commission rates, driving competition among sellers. Most closely related to my empirical approach, Yu (2024) examines the welfare effects of sponsored product advertising on Amazon using a structural discrete choice model with search frictions, finding that eliminating sponsored ads could increase consumer surplus by 13% and seller profits by 14%, though these gains are sensitive to how platforms adjust their fee structures in response. My paper builds on this framework by incorporating platform ranking algorithms and seller delivery decisions into the structural model, as well as consumers choosing across platforms, allowing me to separately identify the role of algorithmic bias from genuine quality differences.

The second strand concerns vertical integration in platform and two-sided market settings, where almost all existing evidence is about platforms favoring their own competing products rather than their own complementary services. The classical literature establishes that upstream firms can foreclose downstream rivals by raising their costs or restricting access to essential inputs (Ordover et al., 1990), and that vertical mergers can harm welfare even absent horizontal market power (Hart and Tirole, 1990). These insights were extended to incomplete-contract settings by Grossman and Hart (1986),

who show that integration affects investment incentives by shifting residual control rights, a mechanism particularly relevant when platforms decide whether to internalize delivery services. In platform settings specifically, Lee (2013) shows that vertical integration and exclusive dealing can foreclose rivals and harm welfare even when the integrated product is of higher quality, since the platform can leverage its control over access to distort downstream competition. Closer to my setting, a smaller theoretical literature models platforms as biased intermediaries that distort rankings to favor an affiliated product or service: de Cornière and Taylor (2014) and de Cornière and Taylor (2019) show that an integrated intermediary has an incentive to bias its recommendations toward its own service even at the expense of match quality, and Hagiu and Jullien (2011) model why intermediaries divert search toward favored options in the first place. Cure et al. (2022), using web-scraped data from the search platform Kayak, document that affiliated online travel agencies receive more prominent placement than equally-priced rivals, a pattern similar to what I document for integrated delivery on Amazon and Mercado Libre, though in a different industry and without a structural model linking prominence to prices. I build on the empirical strategy of Lee (2013) and allow consumers to choose which platform to join based on their expectations of the product choice set they will face on each.

Within this strand, Raval (2022) documents fulfillment bias in Amazon’s ranking algorithm. Li et al. (2025) is the closest theoretical paper, modeling rival platforms that differ in whether they offer an integrated fulfillment service. My paper differs from both by structurally estimating, rather than assuming or simulating, the equilibrium link between ranking bias, delivery adoption, and prices, and by doing so for two platforms that actually compete with one another.

This paper is also connected to an increasingly large literature that tries to understand the effect of ranking algorithms on consumer choice. In a foundational paper, Ursu (2018) shows that ranking algorithms affect what consumers search but, conditional on search, do not affect purchases. Based on this evidence, the model in Section 4 assumes that, conditional on observing a product, its ranking does not enter the utility function. More recent papers use structural methods to understand optimal rankings from the platform or consumer welfare perspective (e.g. Compiani et al. (2024), Derakhshan et al. (2022)) and measure whether ranking algorithms are biased toward certain characteristics, e.g. delivery method, beyond what is optimal for matching (Reimers and Waldfogel, 2023). I follow some of these approaches to understand the platform’s ranking decisions and to tease out the effect of the delivery method on rankings, conditional on other observable and unobservable product characteristics.

The fourth strand focuses specifically on Amazon’s competitive conduct. Zhu and Liu (2018) examine Amazon’s entry decisions into third-party product markets, finding that Amazon targets more successful products and that its entry discourages affected third-party sellers while reducing prices for consumers. Farronato et al. (2023) document that Amazon-branded products appear 30 to 60% higher in search results compared to third-party products with similar observable characteristics, providing reduced-form evidence of self-preferencing consistent with my structural estimates. Dubé (2022) offers a contrasting view, arguing that Amazon’s private label strategy resembles stan-

dard supermarket private label practices rather than anticompetitive foreclosure. More recent structural work, including Musolff and Kwok Hao (n.d.), Hartzell and Haupt (2024), Lam (2021), and Chen and Tsai (2024), develops increasingly rich search-and-ranking models on Amazon to separate algorithmic bias from genuine quality differences. This literature is, by construction, about a single platform: because Amazon is the marketplace with the most accessible seller and ranking data, these papers can speak to how self-preferencing redistributes surplus between a platform and its own sellers, but not to how it shapes competition between platforms. A related limitation is that Amazon does not disclose unit sales, so most of this literature either relies on prominence or click-through measures in place of sales (Farronato et al., 2023), or infers demand from Amazon’s Best Seller Rank under an assumed Pareto relationship between rank and unit sales, an approach dating back to Chevalier and Goolsbee (2003) and used throughout more recent structural work (Zhu and Liu, 2018, Musolff and Kwok Hao, n.d., Chen and Tsai, 2024, Lam, 2021). I instead observe monthly sales buckets directly reported in search results on both platforms, which lets me estimate demand without relying on a parametric mapping from ranks to sales. This paper contributes to this literature by providing the first structural estimates of algorithmic bias in an emerging market context, using variation across two competing platforms to separately identify the role of rankings from other sources of platform market power.

The fifth strand is the theoretical literature on competition between two-sided platforms, which motivates bringing two platforms into the same empirical framework rather than studying either in isolation. Armstrong (2006) and Rochet and Tirole (2003) show that platforms compete for users on one or both sides of the market and that the resulting market power depends on how easily users can multihome, and Armstrong and Wright (2007) formalize the “competitive bottleneck” case in which a platform retains market power over one side even under intense competition for the other. This literature predicts that a platform which distorts rankings more aggressively in favor of its own affiliated service should be relatively less attractive to the side bearing the cost of that distortion, and should therefore lose share to a less-distorting rival once regulation levels the playing field. To my knowledge this prediction has rarely been tested against data from platforms that compete directly for the same consumers and sellers.

The rest of the paper is organized as follows. In Section 2, I describe how the platforms operate. Section 3 describes the data collection process and presents descriptive statistics. In Section 4, I develop an equilibrium model that accounts for seller, buyer and platform’s decisions. Section 5 presents the identification and estimation strategy. Section 6 presents the estimation results of the model presented in section 4 and Section 7 presents the counterfactual analysis. The paper concludes in Section 8.

2 Institutional Setting

Amazon and Mercado Libre are both large international retail platforms competing in the Mexican market. They have substantial presence for both consumers and sellers in Mexico. Together they have around 26% market share and close to 20 billion dollars of merchandise a year is sold through these platforms. In addition, according to the

Mexican Competition authority COFECE (2024), 85% of Mexican online sellers use one or both platforms to sell their products. Similarly in the US, Amazon controls close to 40% of all online retail, so the insights from the Mexican market can also guide competition authorities in the US about how to better regulate the platform.

Table 1: Online Retail Market Shares of the Top Platforms in Mexico

Platform	Market Share	GMV (2024)	USA mkt. share
Mercado Libre	15.4%	~\$11 billion	–
Amazon	11.2%	~\$8 billion	37.6%
Walmart	<3%	–	6.4%

Source: Statista (2023, 2024)

Sellers of a large number of products, ranging from consumer electronics and clothes to furniture and groceries, list their products on the platforms. The platforms then collect fees for each product sold through their website and they also offer delivery services for sellers. They also offer a subscription service for consumers where, for a monthly fee, consumers get free delivery of the products delivered by the platforms, as well as other perks such as streaming services.

Table 2: Summary of Fees and Subscription Prices of Amazon and Mercado Libre

	Amazon	Mercado Libre
<i>Integrated Delivery Service:</i>		
Program Name	Fulfilled By Amazon (FBA)	Full
Year Introduced	2015	2013
<i>Seller Fees:</i>		
Warehousing	\$0.35/mo. per cubic dm	\$0.011–\$0.50 per day
Shipping	\$50–\$65 + \$4/lb	\$131–\$3,146
Listing Fee	\$40 USD/month	None
Referral Fee	8–15% (min \$8)	8–16% (min \$25)
<i>Consumer Subscriptions:</i>		
Basic Tier	\$59	\$99
Medium Tier	\$129	\$149

Notes: Shipping fees information for Amazon obtained from <https://sell.amazon.com/pricing> and <https://www.mercadolibre.com.mx/ayuda/24253> for Mercado Libre. Subscription prices obtained from <https://www.amazon.com.mx/amazonprime> and <https://www.mercadolibre.com.mx/suscripciones/melimas>. Amazon referral fees range up to 45% for Amazon-branded devices. All prices in MXN unless otherwise noted.

On the seller side, both platforms charge commission fees and offer optional integrated delivery services, but differ in their entry requirements and operational structure (Table 2). Mercado Libre charges a category-specific commission fee per sale (8%-16%, minimum \$25 MXN) and offers optional advertisement packages (Classic or Premium), with free entry for sellers who opt out of advertising. The platform’s Full service operates as follows: sellers send inventory to Mercado Libre warehouses and pay a

daily storage fee ranging from \$0.011 to \$0.50 MXN depending on product dimensions; when orders arrive, the platform fulfills them and charges a delivery fee between \$131-\$3,146 MXN based on weight and dimensions, plus the standard sales commission. The sellers have an incentive to choose *Full* since the platform promises that it will lead to more visibility for the product and higher sales since consumers trust the delivery service. The platform also actively manages seller inventory decisions through its warehouse management system, alerting sellers when stock levels are low and could run out within days, or when inventory is excessive and could remain in storage beyond 8 weeks (<https://www.mercadolibre.com.mx/ayuda/40074>). This inventory optimization seems to work for sellers: in the data, the average Full product maintains approximately 30 units in stock while generating 242 monthly sales, implying sellers must replenish inventory roughly every 5 days to avoid running out of stock or unnecessary stocking expenses for excessive units.

Amazon’s fulfillment structure imposes higher fixed costs. Sellers subscribe to a selling plan, paying either \$1 USD per item sold (individual plan) or \$40 USD monthly (professional plan), in addition to a category-specific commission (8%-15%, minimum \$8 MXN, with higher rates up to 45% for Amazon-branded devices). Fulfilled by Amazon (FBA) requires sellers to ship products in advance to Amazon warehouses, paying \$0.35 MXN per cubic decimeter monthly for storage. Amazon then handles individual shipments to consumers, charging \$50-\$65 MXN base fee plus \$4 MXN per pound for delivery. These operational differences, particularly Amazon’s mandatory subscription fee and higher minimum delivery costs, create platform-specific entry barriers that discourage multihoming, as sellers must adapt inventory management, pricing strategies, and fulfillment logistics to each platform’s distinct fee structure.

On the consumer side, both platforms offer subscription services that provide free delivery on orders fulfilled through their integrated services, creating indirect network effects that reinforce platform-specific seller strategies. The Amazon Prime subscription costs \$129 MXN monthly (approximately \$7.50 USD) and provides free one-day delivery on FBA orders, plus complementary video and audio streaming services and access to exclusive sales. Mercado Libre’s Meli+ subscription costs \$149 MXN per month, providing one-day free delivery and access to third-party streaming services, including Disney+. These platform-specific fulfillment ecosystems create strong incentives against seller multihoming. In the data, only 21% of products on Mercado Libre and 15% on Amazon appear on both platforms (see Section 3), and even when the same branded product appears on both, the sellers offering those products are different. The combination of platform-specific entry costs (subscription fees, listing optimization, inventory allocation), distinct fulfillment fee schedules, and different consumer subscription bases means that sellers effectively choose one primary platform and optimize their pricing and delivery decisions for that platform’s specific environment. In the model presented in Section 4, I treat seller platform choice as given and focus on their within-platform strategy.

Both platforms employ ranking algorithms designed to present the most relevant

products at the top of search results.¹ Though neither platform publicly discloses full algorithmic details, several factors are known to influence rankings across both platforms. First, product characteristics including pricing, product descriptions, delivery method, and search query relevance based on keywords. Second, user engagement metrics such as click-through rates and conversion rates, along with seller performance metrics including ratings, sales volume, and response rates. For sellers using the integrated delivery, the algorithms also incorporate inventory availability. Third, consumer-specific factors: purchase history affects personalized rankings, making each individual’s search results somewhat unique. Finally, there is a random component in the ranking algorithm, so even if the same customer searches one keyword twice, they will get a slightly different ranking each time.

3 Data and Descriptive Results

I collect data from Amazon and Mercado Libre search results for specific keywords over 108 weeks from May 2023 to May 2025 using web scraping. The data consist of a panel of product search results from both platforms, capturing up to seven pages of results per search. For each product in the search results, I observe its ranking, product name, description, price, delivery type, customer reviews, and seller information including name, reputation score, and total sales. I define a market as a keyword-week pair.

The search keywords in the sample are drawn from Mercado Libre’s weekly ‘most popular’ and ‘highest growing’ trends (<https://tendencias.mercadolibre.com.mx/>), representing the most actively searched product categories on the platform during the sample period. Table 10 in the Appendix lists all keywords collected along with market sizes. The keywords span diverse product categories including electronics, home goods, apparel, and sporting goods, capturing a substantial share of consumer search activity. While the sample does not include all products available on either platform, it targets the product categories with the highest search volume and competitive intensity, precisely the markets where rankings could have the greatest economic significance. The focus on high-traffic keywords also ensures that the sample is representative of the platform segments where consumer search behavior and seller competition are most consequential for market outcomes.

Monthly sales figures are reported by both platforms in buckets of 100 units (e.g., 0-100, 100-200). I use the midpoint of each bucket as the sales measure for analysis. For Amazon products, I corroborate sales and pricing figures using data from [Keepa.com](https://www.Keepa.com), which provides independent tracking of product sales and pricing history. Price measurements incorporate all consumer-relevant costs, including promotional discounts and shipping fees, to capture the effective price consumers face at the time of search. To mitigate potential measurement error from day-specific fluctuations or sampling timing, and to match the seasonality of sales figures, I aggregate observations at the monthly level. This aggregation reduces noise from transient shocks while the extent of my observation period preserves sufficient time-variation to identify the main effects.

¹A product’s ranking is its position in the search results. A lower ranking indicates the product appears higher on the page and therefore has greater visibility.

Table 3: Descriptive Statistics

Variable	Mercado Libre		Amazon	
	Mean	(Std. Dev.)	Mean	(Std. Dev.)
<i>Market Structure:</i>				
Products per Keyword	250.32	(143.96)	351.23	(154.80)
Sellers per Keyword	119.86	(92.75)	77.74	(177.45)
Multihoming products (%)	21.01	(10.16)	17.13	(13.89)
Multihoming sales (%)	9.18	(0.17)	16.78	(0.10)
<i>Product Characteristics:</i>				
Price (MXN)	4,207.97	(3,924.90)	2,564.27	(3,437.87)
Monthly Sales (units)	242.47	(371.91)	106.33	(275.99)
Number of Reviews	330.48	(1,991.29)	223.37	(1,694.97)
Rating (1-5 stars)	4.61	(0.36)	4.67	(1.80)
Commission Rate	15.03%	(4.90)	13.07%	(2.77)
First-Party Delivery (%)	50.66	(49.87)	66.35	(38.31)
FBA/Full Delivery Cost	80.22	(38.58)	97.21	(84.49)
Full Product Stock	30.60	(20.81)	-	-
Observations	186,240		228,910	

Notes: Sample covers 415,150 observations from 12,950 unique seller-product pairs across 50 search keywords tracked over 108 weeks (May 2023-May 2025). Standard deviations in parentheses. Products denotes unique product listings per keyword-week. Sellers counts distinct seller identities offering products within each keyword. Monthly Sales represents average monthly unit sales as reported by platforms. First-Party Delivery refers to Full (Mercado Libre) and FBA (Amazon). All statistics computed at the product-week level.

Importantly for identification, the rankings observed in this study represent the algorithm’s baseline ordering for users with no platform history, since web scraping uses new proxies with no cookies or browsing data. This creates two key advantages for the empirical strategy. First, the observed rankings reflect the platform’s default algorithm without personalization, allowing me to estimate how the algorithm weights product characteristics (price, sales, reviews, delivery method) in Section 5.3.2 using a Rank-Ordered Logit specification. Second, the stochastic component, evidenced by ranking volatility even for products with stable characteristics (I later show that products change ranks an average of 10 times over the sample period) provides identifying variation for separating the causal effect of ranking position on sales from reverse causality, as detailed in the identification strategy in Section 5.1.2.

The platforms charge sellers commission rates that vary by product category and fulfillment fees based on item size and weight. Each product on Amazon and Mercado Libre has a *category tree* indicating the product’s root category (e.g., electronics, clothing), specific subcategories (e.g., speakers, men’s clothing), and finest category (e.g., portable Bluetooth speakers, men’s sport clothing). I collect data on the commission

rate the platform charges in each subcategory and match it to corresponding products.² For fulfillment costs, I match products in the sample with approximate warehousing and delivery costs based on item descriptions and available package size information. In the model, I treat warehousing costs as a part of the item’s delivery cost since I note that sellers keep low stock to sales ratio. Finally, I incorporate aggregate data on platform market shares and market sizes across online retail segments from Statista. The final dataset comprises 415,150 product-week observations across 50 search keywords, covering 12,950 unique seller-product pairs.

Table 3 presents descriptive statistics for the primary sample. The data covers 186,240 product-week observations for Mercado Libre and 228,910 for Amazon. On average, I observe 351 products sold by 77 different sellers per keyword search on Amazon. On Mercado Libre, the average search returns 250 products with approximately 120 distinct sellers.

The platforms show differences in pricing and sales patterns. The average item price for products sold on Amazon is \$2,564 (MXN) compared to Mercado Libre’s \$4,208. Average monthly sales on Mercado Libre are 242 units compared to Amazon’s 106 units. Sales dispersion is higher on Mercado Libre (std. dev. 371.91) relative to Amazon (275.99). Importantly, 66% of products on Amazon use Fulfill By *Amazon* and 51% of products on Mercado Libre use *Full*. Other papers (e.g. Dubé (2022), Farronato et al. (2023)) have looked at Amazon acting as a seller and how it affects product market competition. In Mexico, Mercado Libre never acts as a seller, and in my data, less than 1% of all sales come from products sold directly by Amazon so I abstract from that part of their business and treat products sold by Amazon as if sold by an independent seller in my main results.

Figure 6 in the Appendix presents prices, sales and use of first-party delivery across keyword searches. The use of FBA and Full varies across markets, this variation will be useful when determining the effect of the use of these delivery options on market outcomes. Higher prices in Mercado Libre are correlated with higher prices on Amazon, while sales are slightly negatively correlated. This is evidence that the search results of the two platforms display products that are substitutes.

To assess the extent of seller and product multihoming across platforms, I implement a two-stage matching procedure. First, I match products across Amazon and Mercado Libre by comparing product titles using fuzzy string matching (Levenshtein distance) with a threshold of 85% similarity, supplemented by exact matches on brand names and product descriptions when available. This conservative approach yields an overall match rate of 21.0% for Mercado Libre products and 17.13% for Amazon products (Table 3), indicating limited product-level multihoming. Second, I attempt to match sellers across platforms by comparing seller usernames and store names that sell similar products. However, sellers on both platforms choose their own store identifiers (which can be arbitrary usernames, business names, or individual names) making systematic seller

²Commission rate data are available at <https://vender.amazon.com.mx/precios> for Amazon and <https://vendedores.mercadolibre.com.mx/> for Mercado Libre.

matching infeasible without additional external data sources. I observe no reliable seller matches across platforms in the sample. When the same branded product does appear on both platforms (the 21% overlap), it is typically offered by different sellers or by the same manufacturer using different fulfillment strategies tailored to each platform’s fee structure. This limited multihoming at both the product and seller levels supports the modeling assumption in Section 4 that sellers select a primary platform and optimize their pricing and delivery decisions for that platform’s specific environment, rather than maintaining active presence and coordinating strategies across both marketplaces.

For the main estimation results of the model presented in the following section, I keep the top 50 products of each keyword-week pair as a way of keeping the model computationally trackable. I justify this exclusion criteria with Figure 5, in the Appendix, where we observe that the top 50 products account for 85% of the market share in Mercado Libre and 93% in Amazon on average across all keywords. I then define the remaining products as the outside option in the model. I aggregate the data at the monthly level for estimation. I also drop products that appear in less than 3 consecutive week search results and run additional estimations using alternative exclusion criteria and run estimations on the weekly data. Table 11 in the Appendix presents descriptive statistics for this restricted sample. Most statistics are similar to the full sample, though sales and reviews are naturally higher, since the restriction selects the best-performing products. Importantly, FBA adoption among Amazon’s top 50 products is lower than in the full sample, while Full adoption on Mercado Libre is higher in the restricted sample.

3.1 Product Time Variation

Table 4 documents the frequency of changes in key product characteristics during the observation period. Changes are measured as transitions where a variable’s value at a given month differs from its value at the past month within each product’s time series. I document the distribution of these transitions across the platforms in Figure 7 in the Appendix. The statistics reveal substantial variation in product characteristics over time, indicating that sellers actively adjust prices, fulfillment options, and other strategic variables in response to competitive pressures or platform changes in pricing.

Price adjustments occur frequently on both platforms. Products on Mercado Libre change prices an average of 4.35 times during the sample period, with some products changing prices as many as 18 times. Amazon exhibits similar pricing dynamics, with products averaging 3.44 price changes and a maximum of 17 changes. Changes in the fulfillment method are less frequent than price adjustments, but still occur regularly. On Mercado Libre, products switch between Full and non-Full delivery an average of 1.3 times, with a maximum of 10 switches. Amazon products exhibit slightly more changes in fulfillment, averaging 1.7 changes between FBA and non-FBA, with a maximum of 11 changes. The variation in price and delivery methods across time but within products helps to identify the consumer’s price sensitivity and the perceived quality of the integrated delivery method for each platform.

Sales and market share exhibit moderate variation over time. Products experience

Table 4: Change in Product Characteristics Over Time

Variable	Mercado Libre			Amazon		
	Min	Mean	Max	Min	Mean	Max
Price Changes	1	4.35	18	1	3.44	17
Full/FBA Changes	0	1.28	10	0	1.67	11
Sales Level Changes	2	1.43	22	3	1.66	22
Market Share Changes	2	4.50	23	1	5.27	23
Search Rank Changes	5	9.97	24	3	10.80	24

Notes: Table reports the minimum, mean, and maximum number of transitions in product characteristics during the 108-week observation period. A change is defined as a transition where the variable value at month t differs from its value at month $t - 1$ for each product. Price Changes counts transitions in listed prices. Full/FBA Changes counts switches between first-party fulfillment (Full on Mercado Libre, FBA on Amazon) and seller fulfillment. Sales Level Changes counts transitions between sales buckets. Market Share Changes counts transitions in within-keyword market share. Search Rank Changes counts movements in search result position. Statistics computed at the product level.

an average of 1.4 sales level changes on Mercado Libre and 1.7 on Amazon, measured as transitions between sales buckets. Market share changes occur more frequently, averaging 4.5 transitions per product on Mercado Libre and 5.3 on Amazon, with maximum values reaching 23 changes. The higher frequency of market share changes relative to sales changes reflects both sales changes and compositional changes in the set of competing products within each keyword search.

Search ranking displays the highest volatility among all observed variables. Products on both platforms move across different rank positions an average of 10 times, with some products changing ranks as many as 24 times during the observation period. This substantial ranking variation provides the key source of identifying variation for estimating the determinants of search rank and their effects, as it reflects both algorithm reordering in response to product characteristic changes and exogenous shocks within keyword markets.

3.2 Product Characteristics by Delivery Method and Algorithmic Bias

In this subsection, I document that products using Full and FBA are statistically different from products using third-party delivery, and that they are also treated differently in the platforms' rankings. However, measuring ranking bias is an endogeneity problem: platforms may rank products with higher unobserved quality more favorably, so a product's use of integrated delivery could be correlated with unobserved characteristics that independently affect its ranking. The regression presented below is therefore only a first approximation to the extent of bias, since it does not fully address this concern; the structural model developed in Section 4 explicitly tackles this endogeneity using the identification strategy described in Section 5.1.2.

Table 5 compares product characteristics across delivery methods on each platform. On both platforms, products using integrated delivery are shown in statistically significant better search positions: FBA products rank roughly 9 positions higher than

third-party alternatives on Amazon, and Full products rank almost 8 positions higher on Mercado Libre, a first indication of the ranking advantage documented more formally below. Products using Full and FBA also differ from those using third-party delivery on other dimensions: they are priced higher on both platforms, and on Mercado Libre they generate substantially higher monthly sales. These differences imply that simply comparing rankings across delivery methods to measure algorithmic bias would ignore the effect of prices, sales and reviews on the rankings. On Mercado Libre, for example, the advantage of Full in ranking could come from higher sales. Based on Reimers and Waldfogel (2023), I regress the search rank on delivery status, controlling for product characteristics such as past sales and prices, to have an initial measure of algorithmic bias towards Full and FBA in the platforms’ search rankings.

Table 5: Balance of Product Characteristics: Integrated vs. Third-Party Delivery

Variable	Mercado Libre			Amazon		
	Full	Third-Party	Difference	FBA	Third-Party	Difference
Price (MXN)	4,841.52	3,015.72	1,825.80*** (72.35)	3,059.89	2,702.10	357.80*** (99.96)
Monthly Sales (units)	696.13	253.93	442.20*** (12.89)	111.51	108.41	3.11 (6.95)
Number of Reviews	217.09	193.31	23.78 (53.32)	337.85	224.30	113.55*** (33.28)
Rating (1-5 stars)	4.63	4.66	-0.04*** (0.01)	4.28	4.34	-0.06*** (0.02)
Search Rank	61.30	69.19	-7.88*** (0.67)	59.96	68.85	-8.88*** (0.89)
Observations	94,349	91,891	186,240	151,882	77,028	228,910

Notes: Sample means for products using integrated delivery (Full on Mercado Libre, FBA on Amazon) versus third-party delivery. Difference is Integrated – Third-Party, with the standard error of a Welch two-sample t -test in parentheses below. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 6 presents the results. In the absence of bias, optimal search result rankings should maximize a weighted sum of consumer and seller welfare. Deviations from this welfare-maximizing frontier occur when rankings exhibit bias toward specific product characteristics, such as *FBA* or *Full* delivery services. To test for the presence of such bias, Reimers and Waldfogel (2023) propose regressing product rankings on log prices, market shares, and the suspected bias source; in our case, *Full* or *FBA* delivery status. A coefficient statistically different from zero on the first-party delivery variable constitutes evidence of ranking bias since it implies that similar products are treated differently when using integrated delivery. Our regression results reveal that both *FBA* and *Full* delivery coefficients are negative and statistically significant, indicating that utilization of these services effectively improves product rankings. Specifically, products fulfilled by the platform are systematically positioned in more favorable search positions, suggesting preferential treatment in the ranking algorithm.

Table 6: Reimers-Waldfoegel Measures of Search Bias

	Dependent Variable: Search Rank	
	Amazon	Mercado Libre
Log Price	7.075*** (0.193)	7.679*** (0.138)
Market Share	-81.672*** (13.519)	-153.289*** (23.810)
FBA/Full Delivery	-15.037*** (2.562)	-24.867*** (2.120)
Keyword FE	YES	YES
Time FE	YES	YES
Observations	186,240	228,910
R-squared	0.513	0.524
Adjusted R-squared	0.513	0.524

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The dependent variable is product rank within search results. Log Price is the natural logarithm of product price. Market Share is the product's share of total keyword sales. FBA/Full Delivery indicates whether the product is delivered by the platform's integrated delivery services. All regressions include search keyword and time fixed effects. Standard errors are clustered at the keyword level.

4 Structural Model

In this section, I develop a structural model that captures the strategic interactions among buyers, sellers, and platforms. In the model, two competing platforms set commission fees and delivery fees for sellers, as well as subscription fees for buyers. Sellers choose prices for their products and decide whether to use the platform's first-party fulfillment services to maximize profits. Buyers then select which platform to join, search for a specific keyword, and purchase the product that maximizes their utility. This framework allows me to analyze how vertical integration between platforms and fulfillment services affects market outcomes and welfare.

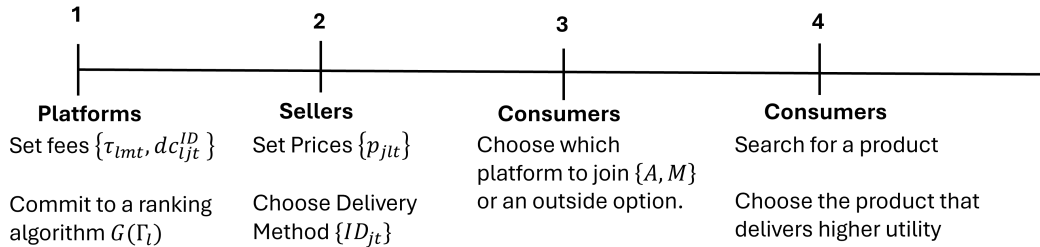


Figure 1: Model Timeline.

Formally, there are $t = 1, \dots, T$ time periods and M markets (keywords) per period.

In each period, two platforms $l \in \{A, M\}$ host sets of sellers $s \in S_{lt}$ who offer differentiated products. In each period there is a continuum of consumers indexed by $i \in I$.

Each time period, the agents play a 4-stage game represented in Figure 1. In stage 1, platforms commit to a ranking algorithm G that maps product characteristics to a distribution of ranks, then set market-specific commission fees τ_{lmt} and delivery fees dc_{ljt}^{ID} for sellers using their integrated delivery services and a subscription fee b_{lt} for buyers. In stage 2, sellers optimally set prices and choose delivery methods given the platforms' fees and ranking algorithm. In stage 3, consumers decide whether to join a platform to search for products or select an outside option. In stage 4, consumers who joined a platform search for a keyword and select the product that maximizes their utility from their choice set.

In the remainder of this section, I describe each stage of the game in reverse order, beginning with stage 4 and proceeding backward to stage 1.

4.1 Buyer Side

4.1.1 Stage 4

Let J_{lmt} denote the set of horizontally differentiated products available on platform l in market m at time t . I index products by j , with $j = 0$ representing an outside good. In stage 4, consumers who join a platform search for a keyword and select the product that maximizes their utility from their choice set. I assume that the platform's ranking algorithm affects consumer choice through search frictions.

Following Yu (2024), I incorporate these frictions in a reduced-form way by assuming that, exogenously, a product in the n -th position can only be examined by a fraction of the consumers in the market and that this fraction is decreasing in ranking. Formally, Let Γ_{lmt} denote the product ranking on platform l , which displays the J_{lmt} products in sequential order. For any position $n \in \{1, 2, \dots, J_{lmt}\}$, define $J_n(\Gamma_{lmt})$ as the consideration set containing all products ranked from position 1 through position n on platform l . Let λ_{nl} represent the fraction of buyers who examine at least the first n products, where the sequence $\lambda_{1l} > \lambda_{2l} > \dots > \lambda_{Nl} > \lambda_{N+1l} = 0$ reflects diminishing visibility as consumers progress down the search results. Consequently, the difference $(\lambda_{nl} - \lambda_{n+1l})$ quantifies the proportion of consumers who examine exactly n products and make their purchase decision based solely on that consideration set. The first few products in search results receive significantly more visibility, creating a substantial advantage for higher-ranked items.

For the main results, I parameterize the fraction of consumers who examine at least the first n products as $\lambda_{nl} = 1/\exp(\beta_l(n-1))$, following Yu (2024). When $\beta_l = 0$, there are no search frictions on platform l and a product's rank does not affect its sales. A larger β_l implies that a greater share of consumers examine only top-ranked products. In the online Appendix I, I relax this assumption and allow λ_{nl} to be non-parametrically identified. Since the estimates are robust to this relaxation, I retain the parametric specification in the main body of the paper.

The indirect utility of consumer i from purchasing product j in market m on platform l is specified as:

$$u_{ijlt} = \underbrace{\alpha_i p_{jlt} + \gamma_l ID_{jt} + \tau_j + \tau_t + \xi_{jlt}}_{\delta_{jlt}} + \epsilon_{ijt} \quad (1)$$

where p_{jlt} represents the price of product j at time t on platform l , with α_i capturing consumer i 's price sensitivity. ID_{jt} is an indicator variable equal to 1 if product j uses the platform's first-party fulfillment service at time t , and γ_l captures the platform-specific utility benefit consumers derive from products fulfilled by the platform (e.g., better customer service). τ_j is a product fixed effect, and τ_t represents time fixed effects. ξ_{jlt} captures unobserved product quality, and ϵ_{ijt} is an idiosyncratic preference shock assumed to follow a Type I extreme value distribution.

The assumption that consumers cannot multi-home, that is, they must choose a single platform for their search reflects the reality that consumers typically concentrate their search activities on a single platform. The mean utility component δ_{jlt} is platform-specific, so the same product may deliver different utility levels depending on which platform it is purchased from due to platform-specific prices, sellers' fulfillment choices, and consumer heterogeneity.

Let $\mathbf{p}_{lmt}$ denote the vector of prices on platform l in market m at time t . The market share of product j on platform l at time t can be expressed as:

$$s_{jlt}(\mathbf{p}_{lmt}, \Gamma_{lmt}) = \sum_{n=1}^{J_{lmt}} 1(j \in J_n(\Gamma_{lmt})) (\lambda_{nl} - \lambda_{n+1l}) \frac{\exp(\delta_{jlt})}{1 + \sum_{j' \in J_n(\Gamma_{lmt})} \exp(\delta_{j'lt})} \quad (2)$$

This equation has an intuitive interpretation: for each consideration set J_n that includes product j , we calculate the probability that a consumer who examines exactly n products chooses product j , weighted by the proportion of such consumers $(\lambda_{nl} - \lambda_{n+1l})$. The indicator function $1(j \in J_n(\Gamma_{lmt}))$ ensures that product j is only considered if it appears in the top n positions. The conditional choice probability follows the standard multinomial logit form, where consumers choose the product with the highest utility among those they have examined.

4.1.2 Stage 3

I follow Lee (2013) and assume that the expected utility for consumer i who selects platform l at time t is given by the platform's mean utility δ_{lt} plus the expected utility from purchasing their preferred product on that platform:

$$u_{ilt} = \delta_{lt} - b_{lt} + E[\max_j \{\delta_{jlt} + \epsilon_{ijt}\} | l] + \epsilon_{ilt}$$

where $\{\epsilon_{ijt}, \epsilon_{ilt}\}$ are independent Type I extreme value idiosyncratic preference shocks for platforms and products. I normalize the utility of the outside option to $u_{i0t} = \epsilon_{i0t}$.

In this state, the expected maximum utility is first taken over all the possible choice sets the consumer could face and then over the idiosyncratic shocks that would make the consumer choose one product in their choice set over the other:

$$E[\max_j \{\delta_{jlt} + \epsilon_{ijt}\} | l] = \sum_{n=1}^{J_{lmt}} \int (\delta_{jlt} + \epsilon_{ijt}) \times \mathbf{1}\{\delta_{jlt} + \epsilon_{ijt} \geq \delta_{j'lt} + \epsilon_{ij't}\}_{j' \in J_n(\Gamma_{lmt})} \times \mathbf{1}(j \in J_n(\Gamma_{lmt})) dF(\epsilon) \quad (3)$$

Consumers join the platform that delivers the highest expected utility or choose the outside option.

4.2 Seller Side

4.2.1 Stage 2

In stage 2, each seller $s \in S_{lt}$ on platform l offers a set of products J_{st} at time t , sets prices p_{jlt} for each product, and decides whether to utilize the platform's first-party fulfillment services. Let $ID_{jt} = 1$ if product j is fulfilled by the platform's integrated delivery service and $ID_{jt} = 0$ otherwise.

I assume that sellers do not know the exact ranking their products will receive in period t but know the platform's ranking algorithm. Specifically, let $G : P \times ID \times X \rightarrow \Delta(\Gamma)$ denote a ranking function that maps product prices, delivery methods, and other observable characteristics to a distribution over positions in the search results. The sellers know G but do not know their exact position in the rankings since G is a stochastic function. Sellers then form an expectation of the market share of product j given the ranking algorithm as follows:

$$s_{jlt}^e(\mathbf{p}_{lmt}) = \int s_{jlt}(\mathbf{p}_{lmt}, \Gamma_l) dG(\Gamma_l) \quad (4)$$

The expected profit function for seller s on platform l is then specified as:

$$\pi_{sl} = \sum_t \sum_{j \in J_{st}} [(1 - \tau_{jmt})p_{jlt} - mc_{jt} - dc_{ljt}^{ID} ID_{jt} - dc_{jt}^{tp}(1 - ID_{jt})] M_t s_{jlt}^e(\mathbf{p}_{lmt}) - FC_s$$

where mc_{jt} denotes the marginal production cost of the product j at time t , dc_{ljt}^{ID} is the cost per-unit of delivery charged by the platform when using the integrated service, dc_{jt}^{tp} represents the cost per-unit when using third-party delivery services, M_t captures the size of the market t , and FC_s represents the fixed costs incurred by the seller s .

The supply-side model captures the key trade-off sellers face: first-party fulfillment may entail higher costs but could lead to preferential ranking by the platform, which in turn increases the expected market share s_{jlt}^e .

4.3 Platforms

4.3.1 Stage 1

As is standard in the literature (e.g., Gutierrez (2021); Castillo (2025); Yu (2024)), I assume that the platform sets commission rates τ_{jmt} and delivery prices dc_{jlt}^{ID} to

maximize a combination of short-term revenues, consumer surplus, and seller profits.

$$\max_{\tau_{jmt}, dc_{jlt}^{ID}, G(\Gamma_{lmt})} \sum_j M_t(\tau_{jmt} p_{jlt} + dc_{jlt}^{ID} ID_{jmt}) s_{jlt}^e(\mathbf{p}_{lmt}) + \mu_l^C CS(\tau, dc^{ID}) + \mu_l^P PS(\tau, dc^{ID}) \quad (5)$$

where $\mu^C, \mu^P \geq 0$ represent the weights that the platform assigns to consumer and seller surplus, respectively.

4.3.2 Ranking Algorithm

The platform also commits to a ranking algorithm $G : P \times ID \times X \rightarrow \Delta(\Gamma)$ which denoted the ranking function that maps product prices, delivery methods, and other observable characteristics to a distribution over positions in the search results. While the platform could in principle choose from a vast space of ranking algorithms, I impose structure by assuming products are ranked according to a Rank-Ordered Logit (ROL) model (Beggs et al. (1981)).³ This specification is flexible yet tractable, allowing the ranking to depend on multiple product characteristics.

In this model, the platform assigns a value or index to each product that is a linear function of its characteristics (p_j, FPF_j, X_j) plus a Type I extreme value unobserved shock:

$$value_j(p_j, FPF_j, X_j) = \beta_p p_j + \beta_f FPF_j + \beta_X X_j + \epsilon_j$$

The platform then ranks products based on the realization of this value, with the highest value ranked first. The probability that product j is ranked before products $\{1, \dots, k\} \subset J$ is:

$$Pr(value_j > \max\{value_1, \dots, value_k\}) = \frac{\exp(value_j)}{\sum_{i=1}^k \exp(value_i)} \quad (6)$$

The probability of observing a specific ranking can be written as the product of such terms, representing a sequential process in which the platform selects the most preferred alternative, then the most preferred alternative among the remaining products, and so on.

5 Identification and Estimation

5.1 Demand Side

5.1.1 Product and Platform Utilities

The product mean utilities δ_{jlt} are identified from observed market shares, where a version of the Berry (1994) and the Hotz and Miller (1993) inversion theorem applies to equation (2). For the platform utilities, we have:

$$\delta_{lt} - p_{lt} + E[\max_j \{\delta_{jlt} + \epsilon_{ijt}\} | l] = \ln(s_{lt}) - \ln(s_{0t}) \quad (7)$$

³An alternative approach assumes the platform uses a linear ranking algorithm. I conduct a robustness test under that assumption and present the results in the online Appendix II.

This allows for identification of δ_{lt} by recovering the expected maximum utility from the logit formula and subtracting it from the left-hand side of equation (7).

5.1.2 Search Frictions and Endogeneity

The key primitives to estimate are consumers' search frictions, captured by $\lambda_{nl} \sim \exp(\beta_l)$, and consumers' price sensitivity α . Estimating the causal effect of a product's rank on its sales faces an endogeneity problem: the unobserved demand shock ξ_{jlt} in equation (2) may be correlated with the product's rank r_{jlt} , as platforms likely rank products with higher unobserved quality more favorably.

To address this endogeneity, I follow Yu (2024) and assume that the unobserved demand shock ξ_{jlt} follows a first-order autoregressive process:

$$\xi_{jlt} = \rho_l \xi_{jl,t-1} + \eta_{jlt}, \quad \eta_{jlt} \sim \text{i.i.d.}, \quad \mathbb{E}(\eta_{jlt}) = 0, \quad \eta_{jlt} \perp \xi_{jl,t-1} \quad (8)$$

The identification strategy relies on two key assumptions:

$$r_{jl,t-1} \perp \eta_{jlt} \quad (9)$$

$$\text{Corr}(r_{jl,t-1}, r_{jlt} | \xi_{jlt}) \neq 0 \quad (10)$$

Assumption (9) states that when the platform determines a product's rank, it does not anticipate future demand innovations. Assumption (10) requires that the platform's ranking algorithm considers sales performance over an extended period, not just current sales. Under these assumptions, lagged rank $r_{jl,t-1}$ is correlated with current rank r_{jlt} but orthogonal to the current demand innovation η_{jlt} , providing a valid instrument for identifying the causal effect of rank on sales.

The intuition for identification is as follows. Consider first a product with constant rank over time: fluctuations in its sales reveal the persistence of demand shocks, identifying ρ_l . Now consider products with the same current demand shock ξ_{jlt} but different shock histories: these products will have different current rankings due to their different lagged shocks. Since lagged ranks are orthogonal to current innovations by assumption (9), variation in lagged ranks provides exogenous variation in current ranks, identifying β_l .

5.1.3 Estimation Procedure

Given these assumptions, I estimate β_l and ρ_l for each platform and market using the Method of Moments with the following moment conditions:

$$\mathbb{E} \begin{pmatrix} \eta_{jlt} \xi_{jl,t-1} \\ \eta_{jlt} r_{jl,t-1} \end{pmatrix} = 0 \quad (11)$$

When the panel of products is unbalanced, the moment conditions in (11) can be modified to account for new products that enter the panel or products that drop out as follows:

$$\mathbb{E} \begin{pmatrix} (\xi_{jlt} - \rho_l^{t-\tau_{jl}} \xi_{jl,\tau_{jl}}) \xi_{jl,t-1} \\ (\xi_{jlt} - \rho_l^{t-\tau_{jl}} \xi_{jl,\tau_{jl}}) r_{jl,t-1} \end{pmatrix} = 0 \quad (12)$$

where τ_{jl} is the first period in which product j is observed on platform l .

Call $\theta_D = (\alpha_i, \gamma_l, \gamma_M, \beta_l, \rho_l, \delta_A, \delta_M)$ the vector of demand parameters that characterize the buyer side of the model. The demand estimation proceeds in the following steps for each platform and market:

1. Given β_l , invert observed market shares s_{jlt} to recover mean utilities δ_{jlt} using equation (2).
2. Estimate platform mean utilities δ_{lt} using equation (4).
3. Regress δ_{jlt} on product characteristics, time fixed effects, fulfillment method indicators (ID_{jt}), and price p_{jlt} to recover the preference parameters. I use a two-stage IV estimator for price where the instruments are average prices of similar products in other markets, delivery costs, and available stock of the product.
4. Calculate the residuals as the unobserved demand shocks ξ_{jlt} .
5. Given ρ_l , calculate the demand innovations η_{jlt} using equation (5).
6. Search for parameters (β_l, ρ_l) that minimize the GMM objective function based on the moment conditions in equation (11) or (12).

Note that the identification and the estimation procedure is done market by market, which can lead to biased estimates due to, for example, independent sampling errors in each market. To account for this, I employ an empirical Bayes procedure so that the market specific parameters are shrunk towards their mean value across markets. So, let $\hat{\theta}^m$ be the market-specific parameters. The Bayes updated estimates are given by

$$\tilde{\theta}^m = \frac{\sigma_{\theta}^2}{\sigma_{\theta}^2 + se(\hat{\theta}^m)^2} \hat{\theta}^m + \frac{se(\hat{\theta}^m)^2}{\sigma_{\theta}^2 + se(\hat{\theta}^m)^2} \bar{\theta}$$

5.2 Seller Side

The next step is to estimate the marginal cost and the delivery cost of the products when shipped through a third-party delivery service based on observed product prices and integrated delivery prices. I use a two-stage method similar to Bajari et al. (2007). In the first stage, I estimate the demand parameters and assume that the seller's pricing strategy is optimal under the structural demand function. Then, any perturbation of the prices should yield profits lower than the original under the original prices (Saeedi, 2019). The cost parameters should satisfy the above condition and then I can estimate them by minimizing the following loss function with respect to third-party delivery prices

$$\sum_{s, \text{ perturbations}} (\pi_{s, \text{perturbed}}(\hat{\theta}_D) - \pi_{s, \text{original}}(\hat{\theta}_D))^2 1[(\pi_{s, \text{perturbed}}(\hat{\theta}_D) - \pi_{s, \text{original}}(\hat{\theta}_D)) > 0] \quad (13)$$

Where $\pi_{s,perturbed}(\hat{\theta}_D)$ is the profit function of seller s under the estimated demand parameters $\hat{\theta}_D$ and perturbed prices, $\pi_{s,original}(\hat{\theta}_D)$ is the profit function under the original prices. The estimation procedure is to simulate buyer’s decisions under the original and perturbed policy functions to compute sellers profits and find the cost parameters that minimize the simulated loss function.

5.3 Platform Parameters

5.3.1 Commission Rates and Delivery Prices

I recover the product-specific weights μ^C and μ^P that platforms assign to consumer and seller surplus using the first-order conditions of the platform’s problem with respect to commission rates and delivery prices. Given observed product prices, commission rates, and delivery prices, these first-order conditions provide a system of equations that identifies the weights.

5.3.2 Ranking Algorithm Parameters

For the Rank-Ordered Logit model of rankings, I estimate the parameters β_p , β_f , and β_X using maximum likelihood based on observed rankings across all markets. The ranking value function includes the following variables and their quadratic interaction terms: current prices, lagged market share, time and market fixed effects, lagged number of reviews, a dummy for positive overall reviews (greater than 4.5 stars, lagged), the delivery method of the product, and time-invariant product characteristics including brand, keyword relevance,⁴ and the length of the product description.

6 Estimation Results

Table 7 presents the parameter estimates of the model, and Table 8 presents the ranking-algorithm estimates. The central estimates for this paper concern integrated delivery so I organize the discussion of the results around three questions: how integrated and third-party delivery services differ, how strongly do the platforms’ ranking algorithms favor integrated delivery, and how much does delivery and price competition shape outcomes on each platform.

On the first question, how integrated and third-party delivery services differ? The estimates show that consumers value the integrated delivery: the taste parameter γ^f is positive on average on both platforms, and larger on Amazon (0.46) than on Mercado Libre (0.32), so FBA is perceived as higher quality than Full and they are both better relative to third-party alternatives. Figure 2a shows the distribution of willingness to pay for the integrated delivery as a percentage of the item’s price. Consumers on Amazon are willing to pay 26% more for a product on average if it is delivered by FBA. Similarly, consumers on Mercado Libre are willing to pay around 25% extra on average for a product delivered by Full. Notice however that these averages hide some

⁴Having a lengthy title that mentions several potential keywords is a known strategy among sellers on these platforms to achieve better rankings. I create a dummy variable equal to 1 if the searched keyword is contained within the product name.

Table 7: Parameter Estimates for Mercado Libre and Amazon

(1) Parameter	Mercado Libre			Amazon		
	(2) Min	(3) Mean	(4) Max	(5) Min	(6) Mean	(7) Max
Demand Shock Persistent Comp. (ρ)	[0.0394	0.2478	0.8136]	[0.0160	0.3248	0.8191]
Search Friction (β)	[0.0000	0.0150	0.0208]	[0.0021	0.0508	0.1567]
Consumer Price Sensitivity (α)	[-8.5146	-4.9322	-3.7662]	[-10.6503	-5.1861	-3.0021]
Full/FBA Taste (γ^f)	[-1.4492	0.3227	2.7388]	[-0.5720	0.4689	2.3258]
Consumer Platform Utility (δ_l)	[0.9358	0.9503	0.9569]	[0.9152	0.9390	0.9636]
Own Price Elasticity	[-2.7372	-1.4112	-0.9840]	[-8.7482	-3.8231	-1.3897]
Markup	[0.2356	0.3712	0.8253]	[0.2341	0.3975	0.8123]
Full/FBA Delivery Cost (\$MXN)	[68.99	229.60	386.43]	[60.47	207.44	582.08]
Third Party Delivery Cost (\$MXN)	[20.02	124.91	365.49]	[42.31	105.98	387.07]
Platform Seller Weight (μ_l^P)	[0.0305	0.5701	2.0173]	[0.0094	1.0377	9.2838]
Platform Consumer Weight (μ_l^C)	[0.0127	1.2033	14.3871]	[0.0023	0.3378	2.9456]

Notes: The table shows the mean parameter estimate and the range of the estimates across 50 keywords analyzed tracked over 108 weeks (May 2023–May 2025). The full results of the estimation, along with standard error for all markets are in Table 12 in the Appendix.

heterogeneity, with estimates ranging from -25% to 75% on Amazon and -50% to 125% on Mercado Libre.

FBA and Full, however, are costly to provide for sellers: integrated delivery costs 84% more on average than third-party alternatives (\$208-230 MXN versus \$105-125 MXN, Table 7). Figure 2b shows the delivery cost as a fraction of the item’s final price for products delivered by Full, FBA and third-parties. The average cost ratio when using third party delivery is around 16% while Full is 21% and FBA closer to 30%. The results indicate that both platforms considerable market power in the delivery segment they intermediate.

To understand why sellers would adopt the integrated delivery despite the increase in cost we need to note that both platforms exhibit substantial search frictions. Figure 2c shows that 62% of Amazon consumers consider only the first 20 products, the median consumer examines only the top 15 products, 22% of Mercado Libre consumers stop searching within the first 20 products and a median search depth is around 50 products. So, sellers would adopt Full or FBA if it leads to higher visibility, and therefore higher sales, on the platform.

So, how strongly do the platform’s ranking algorithms favor their integrated delivery services? The Rank-Ordered Logit (ROL) model of platform rankings, is reported on Table 8, a positive coefficient implies a desirable characteristic for the platforms when ranking the products.⁵ The results show that, controlling for price, past sales, and reviews, products using Full or FBA receive a higher ranking value on both platforms. Figure 2d, translates the positive coefficients on Table 8 into ranking positions: Full delivery moves a product up 6 positions on Mercado Libre, and FBA moves a product up 4 positions on Amazon, on average relative to an otherwise identical product using third-party delivery, with an advantage of up to 30 to 40 positions in some markets.

⁵The model fits the data relatively well, in Figure 9 in the Appendix, I show the observed and predicted probability each ranking.

Table 8: Ranking Algorithm Parameter Estimates

Variable	Mercado Libre		Amazon	
	Coef.	SE	Coef.	SE
Price & Delivery:				
Log(Price)	0.061*	(0.033)	0.021*	(0.010)
Full/FBA Delivery	0.660**	(0.293)	0.712***	(0.262)
Log(Price) $\times$ Full/FBA	-0.054	(0.038)	0.112***	(0.037)
<i>Product Performance:</i>				
Relevance Dummy	0.042	(0.075)	0.384***	(0.055)
Log(Monthly Sales _{<i>t-1</i>})	0.164***	(0.015)	0.460***	(0.026)
Good Reviews _{<i>t-1</i>} (>4.5)	0.348***	(0.089)	0.481***	(0.037)
Number of Reviews _{<i>t-1</i>}	-0.064*	(0.034)	0.130***	(0.035)
Good Rev. _{<i>t-1</i>} $\times$ N Reviews _{<i>t-1</i>}	0.064*	(0.034)	-0.130***	(0.035)
Keyword Fixed Effects	Yes		Yes	
Time Fixed Effects	Yes		Yes	
Observations	35,536		30,099	
Log-Likelihood	-29,439		-39,676	

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variable is product rank within search results, estimated using the Rank-Ordered Logit (ROL) model described in Section 5.3.2. All regressions include keyword fixed effects. Standard errors are clustered at the keyword level. Estimated on 50 keywords analyzed tracked over 108 weeks (May 2023–May 2025).

The ROL model of platform rankings also implies that on both platforms higher sales, better reviews and, importantly, a higher price than similar alternatives gives sellers a higher probability of being shown in better rankings.⁶ Since demand is price elastic on both platforms (somewhat more so on Amazon where the mean own-price elasticity is -3.82 versus -1.41 on Mercado Libre), this is evidence that both platforms want to show products that maximize their revenue, as opposed to consumer surplus. Sellers on both platforms have two incentives to increase prices: better rankings and more revenue per sale, and an incentive to decrease prices to get more sales conditional on consumers seeing their product. So, the algorithms simultaneously reduce the visibility of third-party delivery and of cheaper product alternatives.

So, how much does delivery and price competition shape outcomes on each platform? The estimates of how much does Amazon and Mercado Libre favor their integrated delivery are small relative to other estimates found in the literature and that search frictions, specially on Mercado Libre, are also somewhat small. However, consumer price sensitivity is larger than studies that focus on the US, consistent with consumers in Mexico having lower income, and there are large differences in cost of providing the integrated delivery services compared to third-parties. Therefore, the main harm to consumers from vertical integration in this setting comes from the increased costs of integrated delivery and not the matching distortions as in earlier self-preferencing papers (Farronato et al. (2023), Chen and Tsai (2024), Yu (2024)).

Finally, the two platforms differ not only in how they rank products, but in how they weigh the parties on each side of the market. The platform-consumer and platform-seller weights recovered from the first-order conditions (Table 7, rows 10-11) show that

⁶Figures 10 and 11 show the estimated rank effect of using FBA/Full and decreasing the price of an item by 1%.

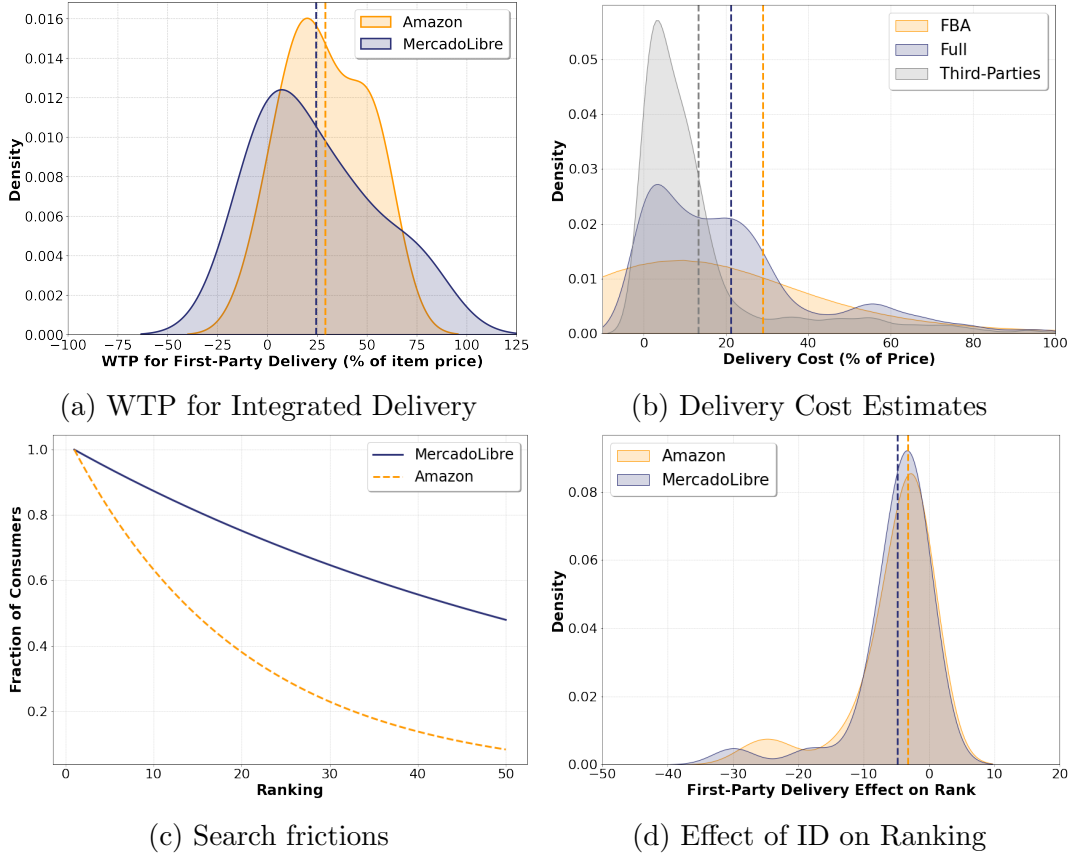


Figure 2: Estimation Results

Note: Notes: Panel (a) shows the kernel density of consumers' willingness to pay for integrated delivery, expressed as a percentage of the item's price, recovered from the estimated taste parameter γ_l^f divided by the price sensitivity α_l for each platform and keyword. Panel (b) shows the kernel density of the estimated delivery cost as a percentage of the item's price, separately for products delivered by FBA, Full, and third-party providers. Panel (c) shows the estimated fraction of consumers who examine at least the first n products for each platform. Panel (d) shows the kernel density of the estimated effect of using integrated delivery on ranking position, relative to an otherwise identical product using third-party delivery. Dashed vertical lines indicate the cross-market mean. Estimates are based on 50 keyword markets tracked over 108 weeks (May 2023–May 2025).

Amazon assigns a higher weight to seller surplus than Mercado Libre does. Combined with the ranking-position asymmetry documented above, this suggests Mercado Libre's algorithm currently extracts more surplus through delivery-based ranking distortions than Amazon's does. This asymmetry is the basis for the counterfactual comparisons in Section 7: because the platforms do not start from the same degree of ranking bias, a common regulatory standard should be expected to affect them differently.

7 Counterfactual Exercises

I conduct two sets of counterfactual exercises. The first asks what happens if both platforms are barred from favoring products that use their own integrated delivery service in rankings. I present the resulting welfare and market-outcome changes through three progressively less restrictive equilibrium concepts, moving from a pure re-ranking exercise to a full equilibrium in which both sellers and platforms reoptimize. The second counterfactual asks what happens if Mercado Libre's integrated delivery service is separated from the platform while Amazon's ranking algorithm and vertical relation

with delivery is left untouched. This counterfactual allows me to examine what is the effect of platform competition on self-preferencing and on the market power of the platforms on the delivery market.

7.1 Unbiased Ranking

I conduct a counterfactual exercise to assess an intervention on the platforms' ranking algorithms aimed at reducing their market power in the delivery segment. In the counterfactual, platforms cannot favor products using their integrated delivery services, which I implement by setting the coefficient on Full/FBA delivery to zero in the Rank-Ordered Logit ranking model (Table 8). The ranking algorithms can still favor higher prices, so price competition under this intervention is softer than it would be under a benchmark in which products are ranked purely by expected consumer utility; I report that benchmark, together with a random-ranking and a no-search-friction benchmark, in the online Appendix III. Under the unbiased-ranking intervention, sellers have weaker incentives to adopt platform-integrated delivery, since doing so no longer buys additional visibility; the only remaining reason to adopt it is its higher quality relative to third-party alternatives.

I report results under three equilibrium concepts that isolate the matching, pricing, and platform-fee margins of the intervention. The first, the *matching effect*, is a partial-equilibrium exercise in which seller prices and delivery choices are held fixed at their baseline values, so that welfare changes arise purely from consumers facing a different, unbiased ordering of the same choice set. The second, the *seller equilibrium*, allows sellers to reoptimize prices and delivery method given the new ranking rule, holding platform commission and delivery fees fixed at baseline. I recover a simulated equilibrium: I simulate the ranking distribution implied by the new algorithm and allow one seller at a time to set its profit-maximizing price, holding all other sellers fixed, then iterate across sellers and update platform market shares until both prices and market shares converge. The third, the *platform equilibrium*, additionally allows platforms to respond to the intervention by adjusting the commission rate paid by sellers; since platforms can no longer charge indirectly for visibility through delivery fees, they may raise commissions on all products uniformly to recoup part of the lost rent. The first two equilibrium concepts assume that the platforms do not respond to the interventions. However, platforms might respond to the new rankings by adjusting other parts of their business, which could dampen the intended effects of the policy so the final counterfactual should be read as the policy relevant exercise.

Figure 3 plots the percentage change in market outcomes across the three equilibrium concepts of the counterfactual relative to the baseline, Table 13 in the Appendix reports the corresponding levels and percentage changes for each panel. Total surplus does not move monotonically across the three equilibria: it rises 6.75% under the *matching effect*, falls back to 3.82% in the *seller equilibrium*, and recovers only partially to 4.56% in the *platform equilibrium*. This pattern arises because the *matching effect* holds prices and delivery choices fixed at values that were optimal under the old, biased ranking, so its 6.75% figure isolates the pure gain from consumers facing a better choice set, evaluated away from the general equilibrium of the game equilibrium. Once

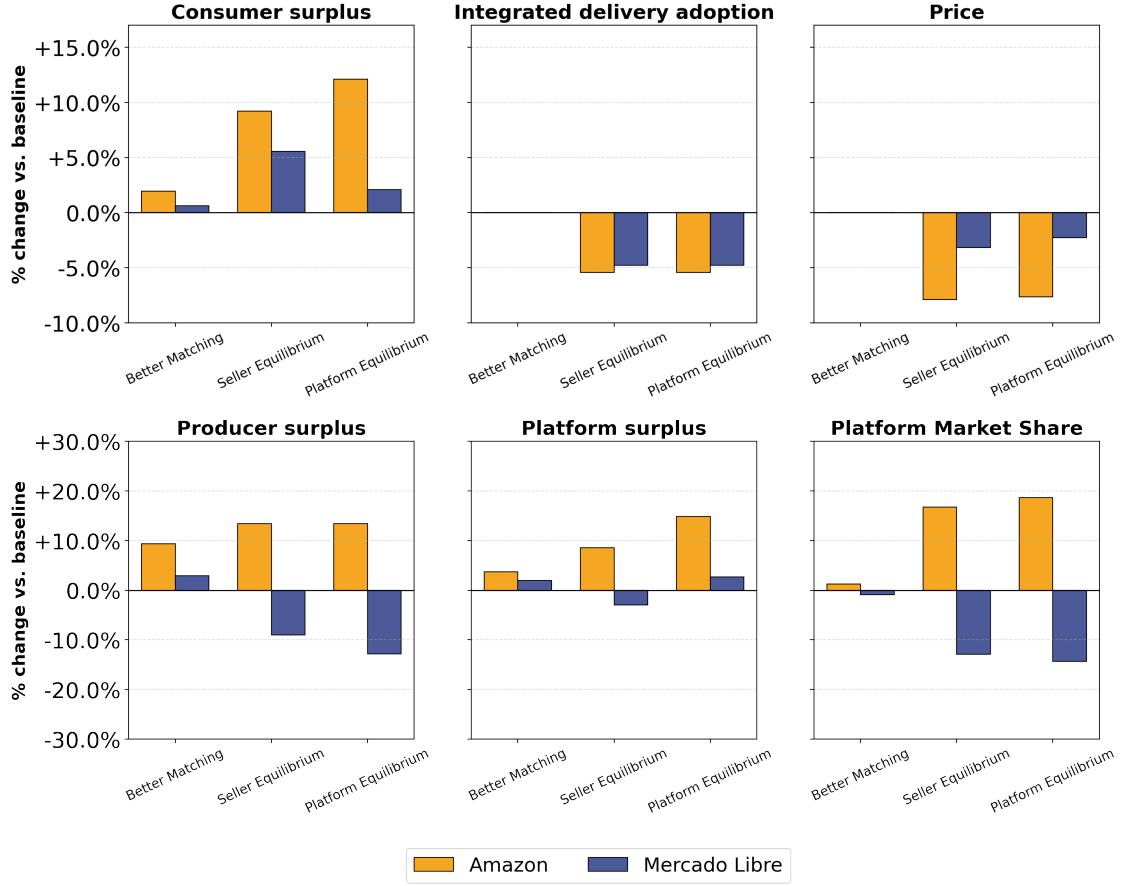


Figure 3: Welfare and Market Outcomes Across all Counterfactuals

Notes: This Figure presents the percentage change of different welfare measures in three equilibrium counterfactual scenarios - Better Matching, Seller Equilibrium, and Platform Equilibrium - compared to the baseline equilibrium. The changes are measured using average per-transaction surplus. Total surplus is the sum of consumer, producer, and platform surplus across both platforms, weighted by market size and market share. Analysis based on 50 Amazon and Mercado Libre keywords. Generated on 2026-02-23.

sellers actually reoptimize prices and delivery method in response to the new ranking rule, the resulting price and delivery adjustments partly undo this matching gain, and platforms' subsequent fee response claws back only part of the difference. The *platform equilibrium*, not the *matching effect* or the *seller equilibrium*, is therefore the relevant welfare benchmark for evaluating the intervention in equilibrium.

Consumer surplus rises on both platforms under every equilibrium concept, but by increasingly different amounts. Under the *matching effect*, consumer surplus rises 1.96% on Amazon versus 0.62% on Mercado Libre, consistent with Amazon consumers facing higher baseline search frictions and therefore benefiting more from a less distorted ranking. These small consumer surplus gains are also consistent with the relatively small degree of self-preferencing by the platforms. The gap widens in the *seller equilibrium*: consumer surplus rises to 9.19% on Amazon versus 5.57% on Mercado Libre, tracking the platforms' price declines (Panel (c)), which reach -7.89% on Amazon against -3.18% on Mercado Libre, alongside similar-sized drops in integrated delivery adoption on both platforms (Panel (b)): -5.42% on Amazon, -4.79% on Mercado Libre) as sellers move away from more expensive integrated delivery. In the *platform equilibrium*, as

Mercado Libre raises its commission rate by nearly 17% to recoup lost delivery rents, its consumer surplus gain falls back from 5.57% to 2.10%, while Amazon’s consumer surplus continues to rise, reaching 12.09%.

Amazon’s producer surplus, platform surplus, and consumer market share all rise steadily across the three equilibrium concepts (Panels (d)–(f)), reaching +13.4%, +14.8%, and +18.6% above baseline, respectively, in the platform equilibrium. Mercado Libre moves in the opposite direction on all three margins: producer surplus falls 9.0% in the *seller equilibrium* and 12.8% in the *platform equilibrium*, platform surplus falls by up to 4.7%, and consumer market share falls by up to 14.3%. This is the same mechanism discussed in Section 6: because Mercado Libre’s baseline algorithm favors Full more aggressively than Amazon’s favors FBA, it must reorder its results more substantially once both platforms are held to a common ranking standard, and this larger reordering costs it market share and rents that migrate toward the less-biased platform. A regulatory change that looks neutral, since it applies the same rule to both platforms, therefore has real distributional consequences for platform competition.

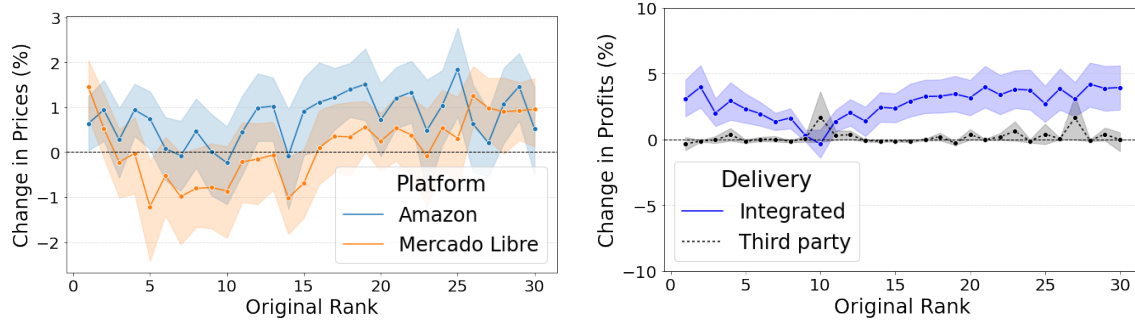


Figure 4: Prices and Profit Changes by Baseline Ranking.

Note: The Figure shows the percentage change in prices and profits under the platform equilibrium counterfactual, plotted against each product’s baseline (pre-intervention) ranking position. The left panel shows the average price change by platform. The right panel shows the average profit change by delivery method, pooling both platforms. Shaded regions indicate 95% confidence intervals.

Figure 4 documents how these aggregate patterns are distributed across products, using each product’s baseline (pre-intervention) rank. On both platforms, price declines are concentrated among products on the top ranks while products on lower rank positions increase their prices on average (left panel). Profit gains are concentrated among sellers who originally used integrated delivery across all ranking positions (right panel). This is consistent with the mechanism above: integrated-delivery sellers capture a disproportionate share of the producer-surplus gains precisely because they no longer have to overpay for visibility through the ranking algorithm, while third-party sellers see comparatively little change.

As discussed in the introduction, two modeling assumptions matter for interpreting these results. First, consumers and sellers single-home in all counterfactuals. If some consumers compare products across Amazon and Mercado Libre, they can partly avoid the effects of ranking bias on either platform, so the consumer surplus gains reported in Table 13 and Figure 4 should be read as an upper bound. Second, the set of sellers and products on each platform is fixed. Under unbiased or utility-based ranking,

sellers using third-party delivery become more visible, which could attract entry and push prices down further than the model predicts. If platforms instead respond to the regulation by raising commission rates, some sellers may exit and reduce the options available to consumers. The net effect of seller entry and exit on the welfare estimates is therefore ambiguous.

7.2 Mercado Libre Divestiture

The unbiased-ranking exercise in Section 7.1 regulates both platforms' algorithms directly. Here I ask a different question: how much of the market power documented in Section 6 is disciplined by competition between Amazon and Mercado Libre, rather than by the design of either platform's own algorithm? To answer this, I consider a counterfactual in which Mercado Libre's Full service is separated from the platform: I set Mercado Libre's delivery prices equal to third-party delivery prices and remove any ranking advantage from using Full, while leaving Amazon's ranking algorithm and FBA service completely unchanged. I allow the platforms to re-optimize their commission fees and the sellers on each platform to re-optimize their prices and delivery methods. This exercise is not intended as a realistic regulatory proposal; no authority has proposed divesting a specific rival's delivery arm. Rather, it approximates what would happen to market outcomes, and to Amazon's own consumers in particular, if Amazon's self-preferencing incentives were no longer disciplined by a rival offering a less-biased alternative, that is, if Amazon's effective market power in Mexico approached that of a monopolist in this segment. Sellers and consumers and both platforms are allowed to respond in equilibrium. Table 9 reports the results.

Mercado Libre's own consumers benefit substantially from losing Full: with delivery repriced at third-party rates, consumer surplus on Mercado Libre rises by 7.73%, even as its average price falls only modestly by 1.20%. This gain comes almost entirely at the platform's own expense. Producer surplus and platform surplus on Mercado Libre fall sharply, by 45.39% and 43.34% respectively, even after Mercado Libre raises its commission rate by nearly 15%, and its consumer market share contracts by 13.04%.

Amazon, meanwhile, gains on every dimension. Its market share rises by 16.92%, and its producer and platform surplus rise by 6.69% and 38.79%, respectively, as consumers from Mercado Libre relocate to Amazon. Consumer surplus on Amazon, however, *falls* by 10.56%, even though FBA prices and quality are held fixed throughout this exercise. With its principal competitor's constraint removed, Amazon behaves closer to an unconstrained incumbent, and its own consumers bear the cost.

Total surplus falls by 4.43%, once Mercado Libre's competitive discipline on Amazon is removed. Unlike the unbiased-ranking exercise in Section 7.1, where regulating ranking bias directly *raises* total surplus, this decline is driven by weaker competition. This 4.43% figure is therefore a useful, if partial, estimate of the value of platform competition in this market, distinct from and additional to the direct value of regulating algorithmic bias documented above. In this exercise the divestiture of Mercado Libre and Full stops disciplining Amazon's incentive to favor its own delivery service and harms consumers.

Table 9: Welfare Analysis: Counterfactual Simulation of Mercado Libre Divestiture

Welfare	Baseline	MeLi Divestiture	% vs Base
Amazon			
Consumer Surplus	\$539.49	\$482.48	-10.56%
Producer Surplus	\$522.77	\$557.78	6.69%
Platform Surplus	\$302.77	\$420.21	38.79%
FBA (%)	53.13%	72.51%	36.47%
Mean Price	\$2,735.88	\$2,599.68	-4.98%
Market Share (%)	34.82	40.72	16.92%
Commission (%)	13.07%	14.87%	13.84%
Mercado Libre			
Consumer Surplus	\$1,499.82	\$1,615.71	7.73%
Producer Surplus	\$885.80	\$483.67	-45.39%
Platform Surplus	\$575.29	\$325.99	-43.34%
Full (%)	0.66	0.00	-100.00%
Mean Price	\$3,201.73	\$3,163.40	-1.20%
Market Share (%)	45.18	39.28	-13.04%
Commission (%)	15.03%	17.26%	14.84%
Total Surplus	\$2,669.10	\$2,550.78	-4.43%

Notes: This table presents the welfare effects of counterfactual scenarios compared to the baseline. Consumer surplus, producer surplus, platform surplus, and total surplus are measured in MXN. Mean price is the average across products in MXN. ID (%) represents the integrated delivery adoption rate. Total surplus is the sum of consumer, producer, and platform surplus across both platforms, weighted by platform market shares. Analysis based on 50 keywords. Generated on 2026-08-17.

Taken together, these results have two implications for competition policy. First, both Amazon and Mercado Libre currently place excessive weight on promoting their own delivery services through rankings: requiring both platforms to rank products without regard to delivery method raises total surplus by 4.56% in the full platform equilibrium, lowers prices, and raises producer surplus. Second, direct algorithmic-bias regulation is not the only lever available, nor necessarily the largest: the divestiture exercise suggests that the mere presence of a competing platform disciplines self-preferencing even absent any change to ranking rules, and that weakening this competitive channel costs total surplus, and Amazon’s own consumers specifically, on a scale comparable to the direct effect of algorithmic bias documented above. This suggests that competition authorities evaluating remedies in two-sided markets should weigh not only how a given platform ranks products, but how much a rival’s presence, or absence, already constrains that ranking in equilibrium.

8 Conclusion

This paper examines the competitive effects of vertical integration between retail platforms and delivery services in Mexico’s online marketplace. Using novel panel data of search results from Amazon and Mercado Libre over two years, I develop and estimate a structural model that incorporates consumer search behavior, platform ranking al-

gorithms, and seller strategic decisions. Platform-delivery integration affects market outcomes through two distinct channels: efficiency gains from improved service quality and market power effects through algorithmic bias in product rankings.

The empirical evidence shows substantial search frictions on both platforms. On Amazon, 62% of consumers consider only the first 20 products and the median consumer examines only the top 15 products, while on Mercado Libre 22% stop searching within the first 20 products and the median search depth is around 50 products. These frictions make ranking position valuable. At the same time, consumers value integrated delivery: they are willing to pay about 26% more for FBA on Amazon and 25% more for Full on Mercado Libre. Yet the integrated services are about 84% more expensive than third-party delivery, and the ranking algorithms systematically favor them beyond these quality differences. Conditional on price, past sales, and reviews, Full moves a product up about 6 positions on Mercado Libre and FBA about 4 positions on Amazon on average. The algorithms also favor higher-priced products, so they reduce the visibility both of third-party delivery and of cheaper product alternatives.

In a counterfactual exercise, platforms are barred from conditioning rankings on the use of integrated delivery. Under the regulation, total surplus rises by 4.56% even after the platforms respond by rising their commission fees by 14-17%. Integrated delivery adoption falls by 5.42% on Amazon and 4.79% on Mercado Libre, while prices fall by 7.65% and 2.27%, respectively. Consumer surplus rises by 12.09% on Amazon and 2.10% on Mercado Libre. The intervention also reallocates consumers across platforms: Amazon's consumer market share rises by 18.60%, while Mercado Libre's falls by 14.34%. Mercado Libre responds to the loss of delivery rents by raising its commission rate, which reduces part of the consumer gain. Because Mercado Libre begins with the more delivery-biased ranking algorithm, the same regulatory standard requires a larger adjustment on Mercado Libre and shifts consumers and rents toward Amazon. A neutral ranking rule can therefore have large distributional consequences across competing platforms.

A second counterfactual aims to argue that platform competition is itself an important constraint on self-preferencing. I separate Mercado Libre's Full service from the platform while leaving Amazon's ranking algorithm and FBA service unchanged. Mercado Libre consumers benefit from the lower delivery cost, with consumer surplus increasing by 7.73%, but Mercado Libre loses 13.04% of its consumer market share. Amazon gains market share and higher profits, but consumer surplus on Amazon falls by 10.56%. Total surplus falls by 4.43%. Thus, removing Mercado Libre's competitive discipline allows Amazon to exercise more market power even though Amazon's own delivery technology and ranking rule are not directly changed. The magnitude of this welfare loss is comparable to the direct welfare gain from eliminating delivery-based ranking bias, highlighting that direct regulation of self-preferencing and preservation of platform competition are distinct but quantitatively important policy margins.

Several limitations of this analysis suggest directions for future research. First, the platform-equilibrium counterfactual allows platforms to adjust commission rates but holds other platform policies, including integrated-delivery prices and subscription fees,

fixed. A richer analysis could allow platforms to reoptimize all of these instruments jointly after a change in ranking regulation. Studying the change in subscription fees would require additional data on the consumer side of the market and a longer observation period, since platforms rarely change their subscription prices. Second, search frictions may themselves be endogenous to the quality of the platform’s ranking algorithm: as ranking improves, consumers may have less incentive to search beyond the top results, so treating search frictions as fixed in the counterfactual abstracts from this additional behavioral response. However, estimating a model of endogenous search would require data on individual search behavior which was not available for this study. Another concern is that the platforms might change the weights in the search ranking after the intervention in a way that we cannot anticipate without placing assumptions on their behavior. Finally, the Mercado Libre separation exercise is intended to isolate the competitive mechanism rather than to model the institutional details of an actual divestiture; incorporating entry and seller platform choice responses would be useful for evaluating structural remedies directly.

Despite these limitations, the results provide evidence that platform-delivery integration creates a misalignment of incentives that distorts search rankings and affects both within-platform efficiency and competition between platforms. Ranking algorithms are a mechanism through which vertically integrated platforms can steer sellers toward their own complementary services, but the extent to which they can profitably do so depends on the competitive alternatives available to consumers and sellers. Policies aimed at platform self-preferencing should therefore consider algorithm design and platform competition jointly rather than as separate sources of market power.

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A Additional Tables and Figures

Table 10: Search Keywords: Category and Sample Coverage

Keyword	Category	Platform(s)	Market size	% of total
aire acondicionado	Home Appliances	Both	109,525	0.97%
andadera	Baby & Kids	Both	30,229	0.27%
arnes electrico moto	Automotive	Both	16,549	0.15%
asus rog phone 9 pro	Electronics	Both	4,346	0.04%
audifonos	Electronics	Both	737,672	6.54%
audifonos con cable	Electronics	Both	735,488	6.52%
banda ejercicio	Sports & Fitness	Both	163,215	1.45%
barra de sonido	Electronics	Both	1,361,740	12.07%
bocina alexa	Electronics	Both	1,364,073	12.09%
cajas de carton	Home & Packaging	Both	61,130	0.54%
carburador tbi	Automotive	Both	9,128	0.08%
chamarras hombre	Apparel	Both	16,100	0.14%
colchon	Home & Furniture	Both	102,484	0.91%
control xbox one	Electronics	Both	343,744	3.05%
esquinero para baño	Home & Furniture	Both	3,092	0.03%
gym shark	Apparel/Sportswear	Both	16,667	0.15%
hielera	Home & Outdoor	Both	60,516	0.54%
horno microondas	Home Appliances	Both	340,000	3.01%
huawei p7	Electronics	Both	17,299	0.15%
ipad	Electronics	Both	37,625	0.33%
iphone 13 reacondicionado	Electronics	Both	9,378	0.08%
kit camaras	Electronics	Both	106,372	0.94%
lamina marmol pvc	Home Improvement	Both	106,775	0.95%
laptop	Electronics	Both	111,140	0.98%
levis 513	Apparel	Both	16,600	0.15%
luces led	Home & Electronics	Both	360,173	3.19%
mens pants	Apparel	Both	16,394	0.15%
mesa comedor	Home & Furniture	Both	609,581	5.40%
mesa jardin	Home & Furniture	Both	609,030	5.40%
minisplit inverter	Home Appliances	Both	59,766	0.53%
motos hot wheels	Toys	Both	105,622	0.94%
mouse inalambrico	Electronics	Both	335,778	2.98%
mueble bano	Home & Furniture	Both	2,842	0.03%
notebook	Electronics	Both	88,558	0.78%
oppo	Electronics	Both	18,024	0.16%
paneles decorativos	Home Improvement	Both	197,181	1.75%
pantalla 32 pulgadas	Electronics	Both	198,956	1.76%
pants de mujer	Apparel	Both	136,582	1.21%
poco m7	Electronics	Both	37,427	0.33%
proyector	Electronics	Both	94,908	0.84%
redmi note 12 pro plus	Electronics	Both	4,730	0.04%
reloj hombre	Accessories	Both	285,034	2.53%
ring camara	Electronics	Both	1,358,323	12.04%
shorts de mujer	Apparel	Both	136,145	1.21%
smart tv	Electronics	Both	92,208	0.82%
smart watch	Electronics/Wearables	Both	163,415	1.45%
teclado pc	Electronics	Both	340,528	3.02%
tenis blancos	Footwear	Both	15,981	0.14%
tenis de mujer	Footwear	Both	135,395	1.20%

Notes: Market size is the combined observed sales on Amazon and Mercado Libre. The % of total is each keyword's share of the total market size across the 50 keywords included in the sample.

Table 11: Descriptive Statistics of the Restricted Sample

Variable	Mercado Libre		Amazon	
	Mean	(Std. Dev.)	Mean	(Std. Dev.)
<i>Market Structure:</i>				
Products per Keyword	45.19	(7.52)	45.37	(3.29)
Sellers per Keyword	37.56	(17.36)	32.37	(33.29)
<i>Product Characteristics:</i>				
Price (MXN)	3544.32	(4624.90)	2580.89	(4584.86)
Monthly Sales (units)	342.88	(431.28)	128.6	(298.1)
Number of Reviews	950.43	(145.05)	452.39	(2023.67)
Rating (1-5 stars)	4.66	(0.3)	4.23	(0.79)
Commission Rate	15.03%	(4.90)	13.07%	(2.77)
First-Party Delivery (%)	61.14%	(48.74)	53.17%	(49.96)
FBA/Full Delivery Cost	149.36	(167.37)	160.28	(33.67)
Full Product Stock	28.35	(20.23)	-	-
Observations	50,050		54,050	

Notes: Sample covers 104,100 observations from 4,069 unique seller-product pairs across 50 search keywords tracked over 108 weeks (May 2023-May 2025), aggregated at the monthly level. Standard deviations in parentheses. Products denotes unique product listings per keyword-month. Sellers counts distinct seller identities offering products within each keyword. Monthly Sales represents average monthly unit sales as reported by platforms. First-Party Delivery refers to Full (Mercado Libre) and FBA (Amazon). All statistics computed at the product-month level.

Table 12: Demand parameter estimates by keyword and platform (all markets)

Keyword	Platform	N	ρ	β	Price sens. α	ID sens. γ
aire acondicionado	Amazon	40	0.318 (1.507)	0.000 (0.000)	-0.229 (0.204)	0.081*** (0.012)
aire acondicionado	Mercado Libre	40	0.646 (1.294)	0.000** (0.000)	-0.236** (0.110)	-0.016*** (0.005)
andadera	Amazon	40	0.305 (1.786)	0.000*** (0.000)	-1.068*** (0.334)	-0.036 (0.034)
andadera	Mercado Libre	40	0.000 (1.166)	0.010*** (0.001)	-4.259*** (0.494)	0.138*** (0.037)
arnes electrico moto	Amazon	30	0.174 (1.295)	0.010*** (0.000)	-1.248 (0.776)	-0.413*** (0.082)
arnes electrico moto	Mercado Libre	30	0.662 (1.515)	0.000 (0.001)	-1.714*** (0.547)	-0.026 (0.024)
asus rog phone 9 pro	Amazon	40	0.000 (0.574)	0.000** (0.000)	-5.253* (2.711)	-0.612*** (0.226)
asus rog phone 9 pro	Mercado Libre	40	0.170*** (0.000)	0.010*** (0.000)	-0.192*** (0.037)	0.441*** (0.083)
audifonos	Amazon	40	0.122 (0.483)	0.012*** (0.000)	-3.210*** (0.875)	0.017 (0.075)
audifonos	Mercado Libre	40	0.000 (0.069)	0.000** (0.000)	-0.156 (0.118)	2.211*** (0.324)
audifonos con cable	Amazon	30	0.000 (0.707)	0.031*** (0.001)	-2.863*** (0.665)	-0.393*** (0.058)
audifonos con cable	Mercado Libre	30	0.657 (1.044)	0.009*** (0.000)	-0.356 (0.309)	1.238 (0.842)
banda ejercicio	Amazon	30	0.256 (0.387)	0.017*** (0.000)	-0.236 (0.290)	0.059 (0.042)
banda ejercicio	Mercado Libre	30	0.141 (1.017)	0.009*** (0.001)	-4.247*** (0.564)	-0.931*** (0.088)
barra de sonido	Amazon	35	0.289 (0.881)	0.009*** (0.000)	-5.924*** (1.724)	1.185*** (0.269)
barra de sonido	Mercado Libre	35	0.647*** (0.026)	0.000*** (0.000)	-2.038*** (0.590)	-0.166*** (0.062)
bocina alexa	Amazon	40	0.329 (1.396)	0.000*** (0.000)	-0.091 (0.671)	0.097 (0.088)
bocina alexa	Mercado Libre	40	0.743*** (0.117)	0.000*** (0.000)	-2.837*** (0.855)	0.284*** (0.085)
cajas de carton	Amazon	40	0.175 (0.972)	0.000 (0.000)	-2.669*** (0.776)	0.070*** (0.010)
cajas de carton	Mercado Libre	40	0.521 (2.633)	0.001*** (0.000)	-0.668** (0.312)	-0.049*** (0.010)
carburador tbi	Amazon	35	0.207 (0.617)	0.000 (0.000)	-0.979** (0.437)	-0.098* (0.053)
carburador tbi	Mercado Libre	30	0.116 (0.715)	0.000 (0.002)	-10.783*** (2.545)	1.583*** (0.238)
chamarras hombre	Amazon	35	0.259 (1.431)	0.000*** (0.000)	-1.115*** (0.355)	-0.031 (0.035)
chamarras hombre	Mercado Libre	35	0.184 (0.827)	0.007*** (0.001)	-4.109*** (0.538)	-0.879*** (0.083)
colchon	Amazon	30	0.465 (0.969)	0.010*** (0.003)	-0.653*** (0.263)	0.175*** (0.023)
colchon	Mercado Libre	30	0.684 (0.525)	0.007*** (0.000)	-0.590*** (0.201)	-0.108*** (0.020)
control xbox one	Amazon	40	0.000 (0.355)	0.000 (0.001)	-0.519 (0.823)	-0.110*** (0.038)
control xbox one	Mercado Libre	40	0.475 (0.999)	0.000*** (0.000)	-0.834** (0.336)	-0.025*** (0.008)
disco duro solido	Amazon	30	0.169 (0.853)	0.010*** (0.001)	-4.723*** (1.204)	-0.558*** (0.154)
disco duro solido	Mercado Libre	30	1.000 (1.667)	0.000*** (0.000)	-0.102 (0.535)	-0.172** (0.070)
esquinero para baño	Amazon	40	0.499 (3.513)	0.001*** (0.000)	-0.450 (0.939)	-0.046 (0.028)
esquinero para baño	Mercado Libre	40	0.155*** (0.002)	0.011*** (0.000)	-2.916*** (0.158)	0.574*** (0.131)
gym shark	Amazon	40	0.000 (0.145)	0.000*** (0.000)	-0.189*** (0.049)	-0.011** (0.005)
gym shark	Mercado Libre	40	0.507 (1.613)	0.000*** (0.000)	-2.333*** (0.366)	0.028 (0.022)
hielera	Amazon	35	0.000 (0.072)	0.000 (0.000)	-0.663*** (0.159)	1.025*** (0.089)
hielera	Mercado Libre	35	0.302 (1.987)	0.002*** (0.000)	-0.752** (0.306)	-0.051*** (0.011)

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Table 12 – continued from previous page

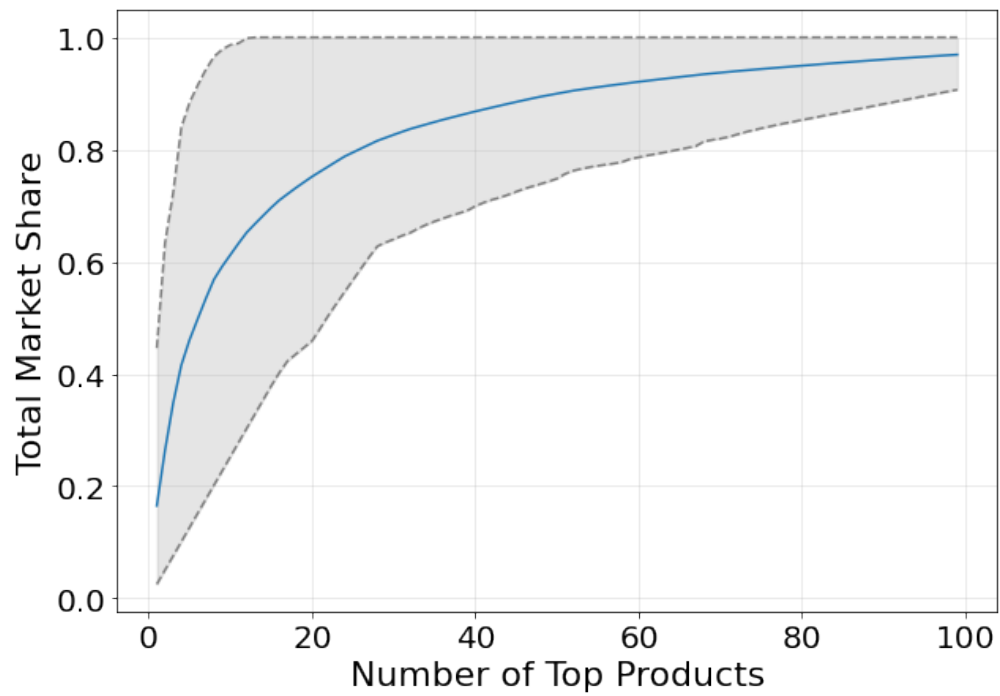
Keyword	Platform	N	ρ	β	Price sens.	FPD sens.
horno microondas	Amazon	30	0.140 (0.683)	0.010*** (0.000)	-7.046*** (0.961)	0.019 (0.178)
horno microondas	Mercado Libre	40	0.292 (0.352)	0.002*** (0.000)	-0.070 (0.176)	-0.232*** (0.073)
huawei p7	Amazon	35	0.002 (0.585)	0.009*** (0.000)	-2.438* (1.424)	-0.243*** (0.080)
huawei p7	Mercado Libre	35	0.495 (2.607)	0.009*** (0.001)	-8.280*** (0.958)	-0.546*** (0.053)
ipad	Amazon	40	0.152 (0.513)	0.005*** (0.000)	-1.608 (1.472)	0.006 (0.035)
ipad	Mercado Libre	40	0.142 (0.283)	0.000*** (0.000)	-0.273*** (0.080)	0.028*** (0.010)
iphone 13 reacondicionado	Amazon	40	0.209 (1.061)	0.000*** (0.000)	-0.924** (0.362)	-0.094** (0.044)
iphone 13 reacondicionado	Mercado Libre	40	0.234 (0.569)	0.002*** (0.000)	-1.808*** (0.664)	-0.023 (0.066)
kit camaras	Amazon	40	0.178 (0.178)	0.001*** (0.000)	-0.065 (0.362)	-0.033*** (0.013)
kit camaras	Mercado Libre	40	0.000 (0.121)	0.000 (0.000)	-1.472*** (0.334)	0.068*** (0.018)
lamina marmol pvc	Amazon	35	0.427 (0.923)	0.010*** (0.001)	-1.047*** (0.357)	0.144*** (0.024)
lamina marmol pvc	Mercado Libre	35	0.159 (0.649)	0.010*** (0.000)	-0.913** (0.381)	-0.235*** (0.031)
laptop	Amazon	30	0.163 (3.949)	0.009*** (0.000)	-5.875*** (0.358)	0.159*** (0.042)
laptop	Mercado Libre	40	0.000 (0.342)	0.000 (0.000)	-0.616*** (0.146)	0.029 (0.019)
levis 513	Amazon	40	0.000 (0.000)	0.000*** (0.000)	-0.019* (0.010)	0.001 (0.001)
levis 513	Mercado Libre	40	0.286 (0.695)	0.009*** (0.001)	-6.414*** (0.637)	-1.155*** (0.097)
lucos led	Amazon	40	0.246 (0.476)	0.001 (0.001)	-0.734*** (0.238)	0.024 (0.021)
lucos led	Mercado Libre	40	0.265 (0.659)	0.000 (0.000)	-0.094 (0.581)	0.025 (0.058)
mens pants	Amazon	35	0.000 (0.124)	0.000*** (0.000)	-0.170*** (0.053)	-0.019*** (0.005)
mens pants	Mercado Libre	35	0.573 (2.164)	0.000*** (0.000)	-2.144*** (0.363)	0.072*** (0.020)
mesa comedor	Amazon	35	0.000 (0.234)	0.000*** (0.000)	-1.450 (1.052)	-0.582 (0.373)
mesa comedor	Mercado Libre	35	0.246 (0.551)	0.003*** (0.000)	-0.467 (0.918)	0.084** (0.035)
mesa jardin	Amazon	30	0.863 (2.942)	0.010*** (0.001)	-0.710** (0.299)	-0.061*** (0.015)
mesa jardin	Mercado Libre	40	0.271 (1.163)	0.002*** (0.000)	-0.164 (1.028)	0.060** (0.029)
mesa jardin†	Mercado Libre	30	0.240 (0.362)	0.004*** (0.000)	-0.639 (0.880)	0.086** (0.035)
minisplit inverter	Amazon	30	0.000 (0.119)	0.001*** (0.000)	-0.785*** (0.199)	-0.113 (0.111)
minisplit inverter	Mercado Libre	30	0.302 (0.346)	0.003*** (0.000)	-1.202*** (0.379)	-0.072*** (0.014)
motos hot wheels	Amazon	35	0.201 (0.565)	0.001*** (0.000)	-0.213 (0.333)	0.002 (0.009)
motos hot wheels	Mercado Libre	35	0.064 (0.470)	0.000 (0.000)	-2.360*** (0.404)	0.099*** (0.023)
mouse inalambrico	Amazon	30	0.000 (0.547)	0.000*** (0.000)	-0.597 (0.482)	-0.175*** (0.031)
mouse inalambrico	Mercado Libre	30	0.668 (6.929)	0.002*** (0.000)	-1.688*** (0.595)	0.017 (0.016)
mueble bano	Amazon	35	0.705 (18.551)	0.000 (0.000)	-0.843*** (0.413)	-0.059*** (0.016)
mueble bano	Mercado Libre	35	0.223 (1.067)	0.000 (0.015)	-25.615*** (5.648)	2.719*** (0.477)
notebook	Amazon	30	0.171 (0.400)	0.011*** (0.000)	-1.022*** (0.255)	-0.286*** (0.057)
notebook	Mercado Libre	30	0.246 (0.748)	0.010*** (0.000)	-0.398 (0.408)	-0.089 (0.055)
oppo	Amazon	40	0.171 (1.086)	0.009*** (0.000)	-2.953** (1.501)	-0.324*** (0.077)
oppo	Mercado Libre	40	0.660 (2.919)	0.000 (0.001)	-1.629*** (0.502)	-0.026 (0.021)
paneles decorativos	Amazon	30	0.129 (0.339)	0.010*** (0.000)	-1.077* (0.604)	0.045 (0.116)
paneles decorativos	Mercado Libre	35	0.146 (0.371)	0.001*** (0.000)	-1.188** (0.472)	-0.156** (0.075)
pantalla 32 pulgadas	Amazon	40	0.187 (0.269)	0.010*** (0.002)	-4.483*** (1.217)	-0.186* (0.097)
pantalla 32 pulgadas	Mercado Libre	40	0.000 (0.151)	0.002*** (0.001)	-1.341** (0.585)	-0.166** (0.084)
pants de mujer	Amazon	40	0.000 (1.951)	0.001*** (0.000)	-0.513 (0.332)	-0.063* (0.034)
pants de mujer	Mercado Libre	40	0.000 (0.199)	0.000*** (0.000)	-10.195*** (3.172)	-0.574*** (0.178)
poco m7	Amazon	30	0.000 (0.884)	0.000 (0.000)	-2.049* (1.093)	-0.027 (0.027)
poco m7	Mercado Libre	30	0.156 (0.883)	0.009*** (0.000)	-1.912** (0.862)	0.185* (0.096)
proyector	Amazon	40	0.627 (1.739)	0.019*** (0.002)	-1.661*** (0.380)	-0.004 (0.059)
proyector	Mercado Libre	40	0.055 (0.373)	0.010*** (0.001)	-1.268 (1.020)	-0.503*** (0.139)
redmi note 12 pro plus	Amazon	40	0.021 (0.082)	0.018*** (0.000)	-0.452 (1.437)	0.101 (0.235)
redmi note 12 pro plus	Mercado Libre	40	0.171 (0.753)	0.009*** (0.000)	-58.134*** (16.580)	-4.291*** (1.402)
reloj hombre	Amazon	40	0.139 (2.613)	0.008** (0.004)	-0.854*** (0.249)	-0.325** (0.142)
reloj hombre	Mercado Libre	40	0.688 (1.918)	0.000*** (0.000)	-1.183*** (0.240)	0.074*** (0.018)
ring camara	Amazon	30	0.239 (1.273)	0.001*** (0.000)	-0.298 (0.246)	0.159*** (0.048)
ring camara	Mercado Libre	30	0.162 (0.337)	0.010*** (0.000)	-14.554* (7.972)	-1.723** (0.838)
shorts de mujer	Amazon	35	0.000 (3.397)	0.001*** (0.000)	-0.353 (0.270)	-0.046* (0.027)
shorts de mujer	Mercado Libre	35	0.000 (0.728)	0.000 (0.000)	-5.890*** (1.488)	-0.331*** (0.091)
smart tv	Amazon	35	0.489 (1.234)	0.010* (0.005)	-0.719** (0.298)	-0.157*** (0.036)
smart tv	Mercado Libre	35	0.041 (0.190)	0.010*** (0.001)	-1.843* (1.081)	-0.386*** (0.115)
smart watch	Amazon	40	0.157 (0.513)	0.011*** (0.000)	-0.755*** (0.238)	0.091** (0.038)
smart watch	Mercado Libre	40	0.135 (0.150)	0.011*** (0.000)	-5.106 (3.669)	-16.784*** (5.308)
teclado pc	Amazon	35	0.000 (0.397)	0.000 (0.002)	-0.319 (0.461)	-0.126*** (0.025)
teclado pc	Mercado Libre	35	0.000 (0.758)	0.000*** (0.000)	-0.412*** (0.132)	0.001 (0.004)
tenis blancos	Amazon	30	0.154 (1.154)	0.009*** (0.000)	-1.112*** (0.203)	-0.047*** (0.016)
tenis blancos	Mercado Libre	30	0.207 (0.393)	0.004*** (0.000)	-2.017* (1.096)	0.302*** (0.082)
tenis de mujer	Amazon	30	0.000 (0.512)	0.002*** (0.000)	-0.443 (0.288)	-0.055* (0.030)
tenis de mujer	Mercado Libre	30	0.144 (0.795)	0.009*** (0.001)	-22.074*** (4.748)	-1.258*** (0.288)

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Rows marked with † duplicate another row for the same keyword and should be checked before use.

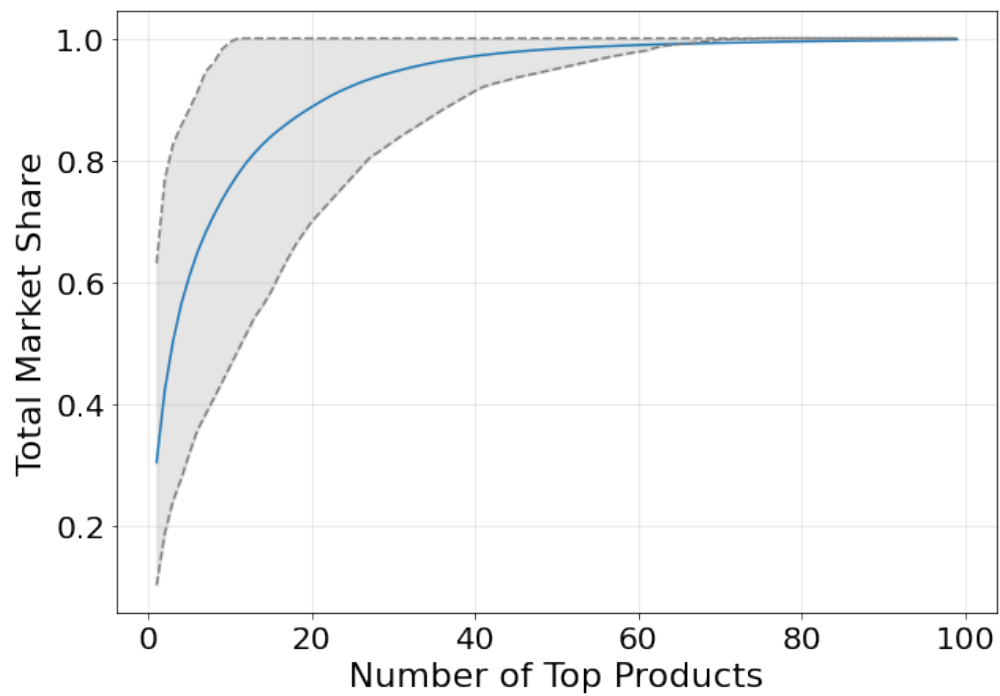
Table 13: Welfare Analysis: Counterfactual Simulation

Welfare	Baseline	Better Matching (% vs Base)	Seller Eqm (% vs Base)	Platform Eqm (% vs Base)
Amazon				
Consumer Surplus	\$539.49	1.96%	9.19%	12.09%
Producer Surplus	\$522.03	9.33%	13.41%	13.38%
Platform Surplus	\$302.77	3.73%	8.58%	14.80%
FBA (%)	53.13	0.00%	-5.42%	-5.42%
Mean Price	\$2,735.88	0.00%	-7.89%	-7.65%
Market Share (%)	34.82	1.21%	16.70%	18.60%
Commission (%)	13.07%	0.00%	0.00%	13.89%
MercadoLibre				
Consumer Surplus	\$1,499.82	0.62%	5.57%	2.10%
Producer Surplus	\$885.80	2.92%	-9.04%	-12.81%
Platform Surplus	\$575.29	1.95%	-2.99%	-4.74%
Full (%)	66.08	0.00%	-4.79%	-4.79%
Mean Price	\$3,201.73	0.00%	-3.18%	-2.27%
Market Share (%)	45.18	-0.93%	-12.88%	-14.34%
Commission (%)	15.03%	0.00%	0.00%	16.99%
Total Surplus	\$2,082.10	6.75%	3.82%	4.56%

Notes: This table presents the welfare effects of counterfactual scenarios compared to the baseline. Consumer surplus, producer surplus, platform surplus, and total surplus are measured in MXN. Mean price is the average across products in MXN. ID (%) represents the integrated delivery adoption rate. Total surplus is the sum of consumer, producer, and platform surplus across both platforms, weighted by platform market shares. Counterfactual columns report percentage changes relative to the baseline. Analysis based on 50 keywords. Generated on 2026-08-15.

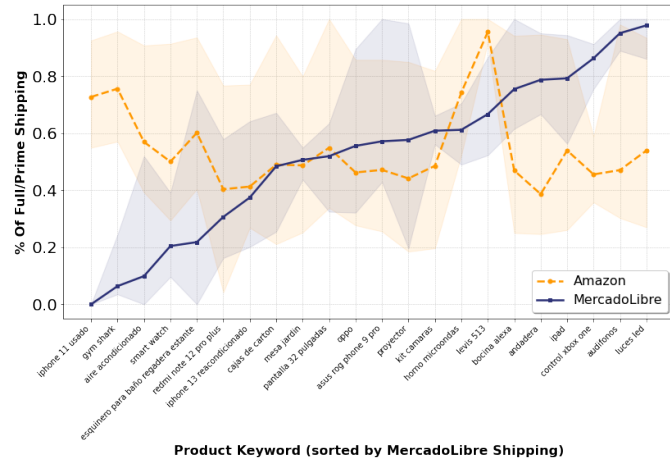


(a) Top Selling: Mercado Libre

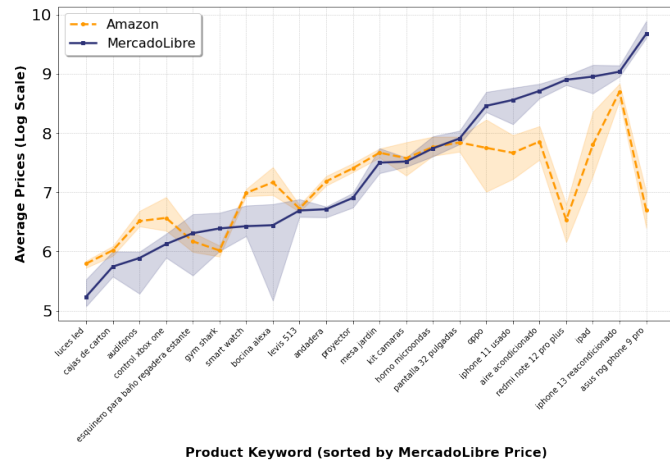


(b) Top Selling: Amazon

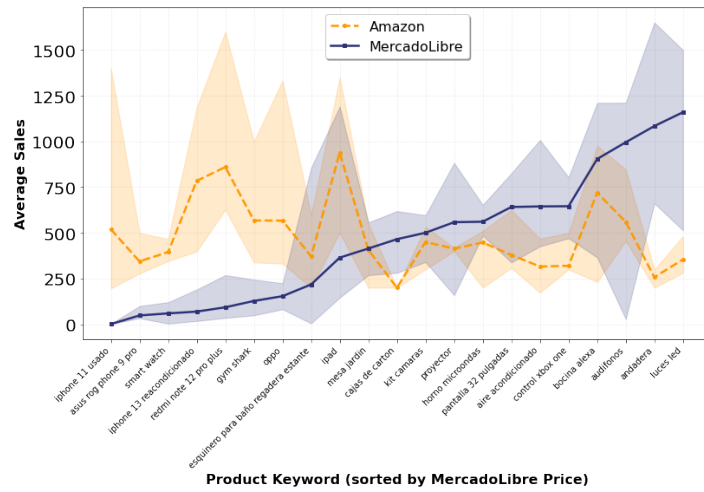
Figure 5: Market Shares of Top Selling Products



(a) Proportion of First-Party Shipping by Market



(b) Average Prices by Market



(c) Average Sales by Market

Figure 6: Statistics by Market on Mercado Libre and Amazon

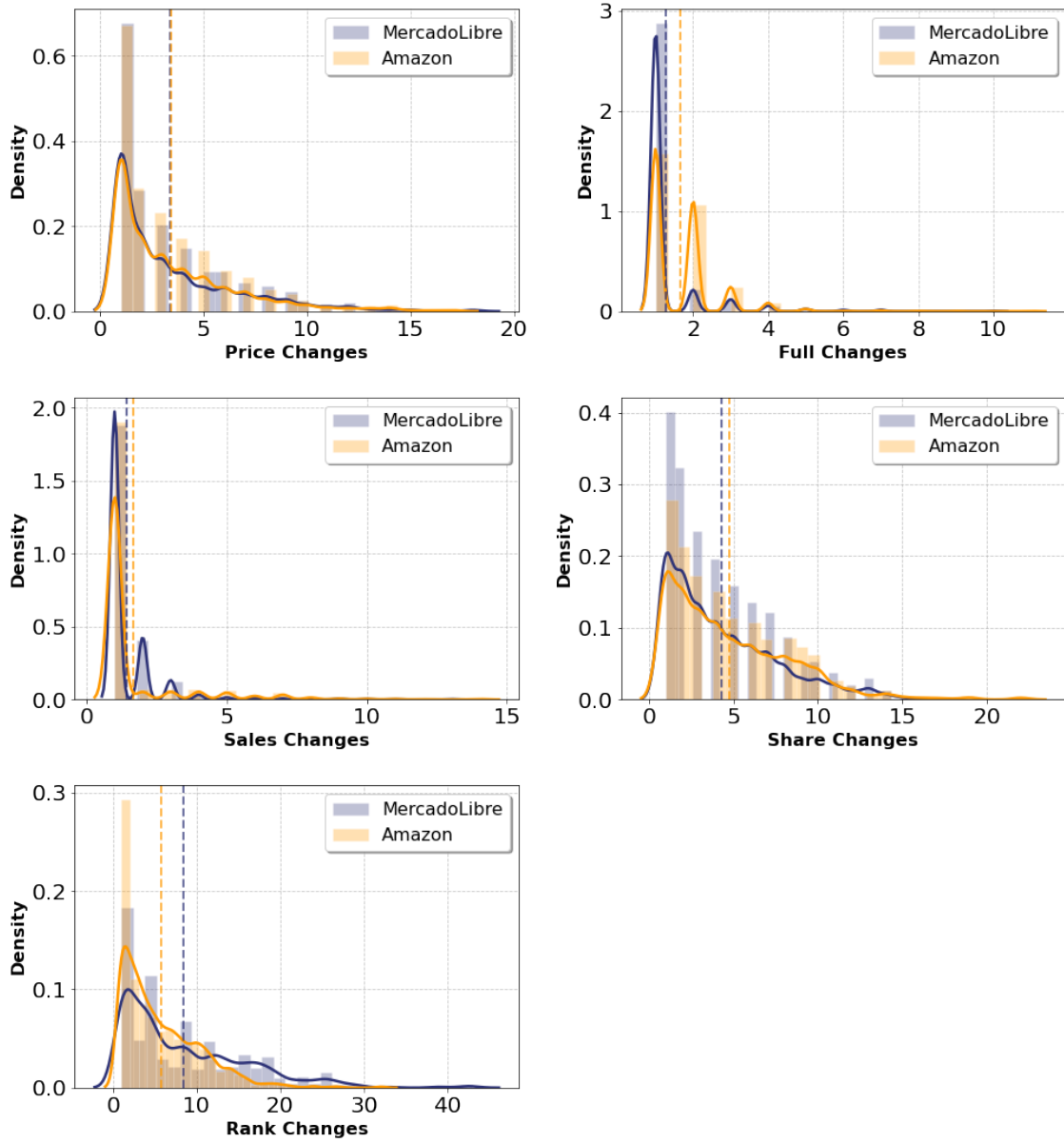
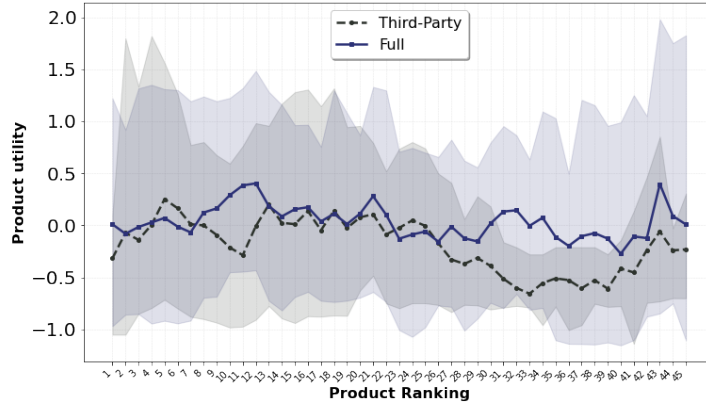
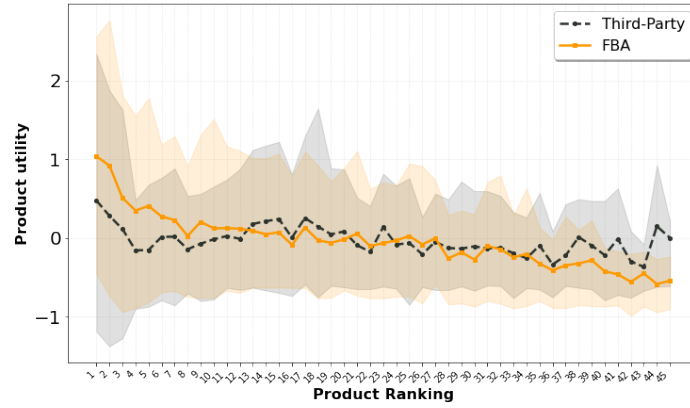


Figure 7: Distribution of Changes in Price, Sales, Ranking and Use of FBA/Full Across Time by Product

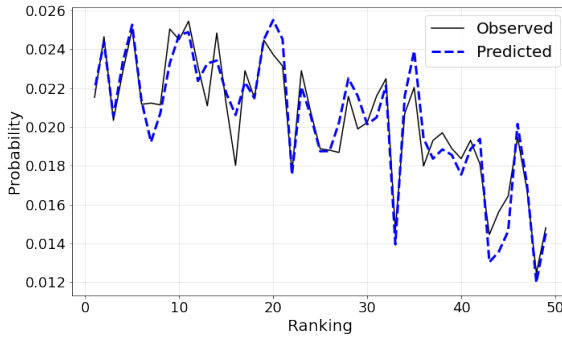


(a) Mercado Libre

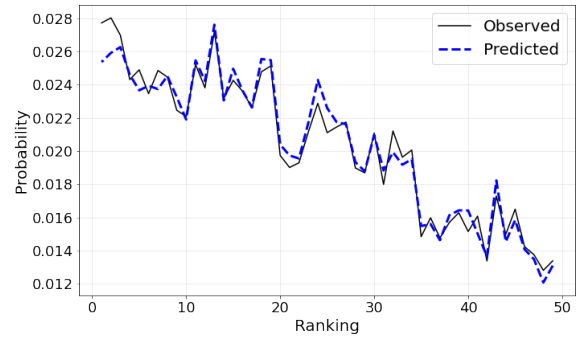


(b) Amazon

Figure 8: Mean product utility by Rank on Mercado Libre and Amazon



(a) Mercado Libre



(b) Amazon

Figure 9: ROL Ranking Probabilities.

Note: The Figure shows the mean observed probability that all products in the sample are placed in a given ranking versus the predicted probability given by the Rank Ordered Logit (ROL) model. A random ranking model would set all probabilities equal for each rank while the ROL matches the trend of the observed probabilities.

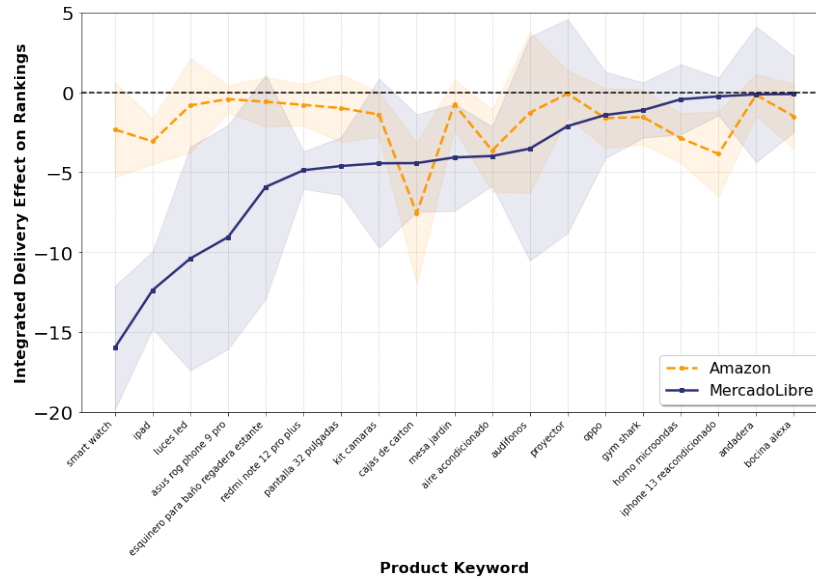


Figure 10: Integrated Delivery Effects on Ranking.

Note: The Figure shows the expected difference in ranking of products using third-party delivery versus those using *FBA* or *Full* across markets using the ROL estimates, all else equal. A negative effect implies that similar products using the integrated delivery systematically appear in lower (better) rankings than those using third party delivery.

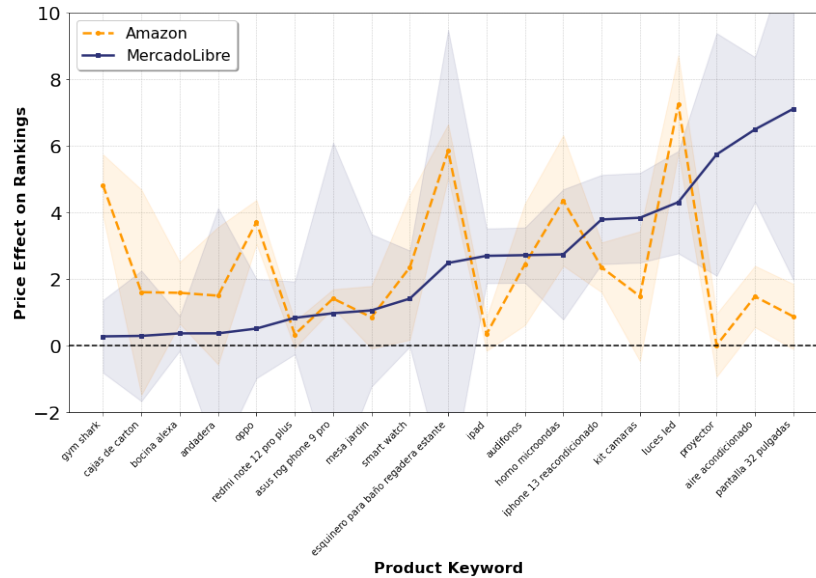


Figure 11: Price Effects on Ranking.

Note: The Figure shows the expected change in ranking when products decrease their prices by 1%, all else equal.